Lecture 10

Advanced Decoding

Last time…

What is decoding?

Decoding is the process of **inferring the underlying representation** from observed neural activity, typically spike times (or continuous data like LFPs or EEG).

Given some neural signal (e.g., spike train), the goal is to **predict a cognitive or behavioral variable (like position, decision, or sensory stimulus)**.

Note: this could be something continuous (like position) or discrete (like decision left or right)

Stats Background: Conditional Probability

Conditional probability is a measure of the probability of an event occurring, given that another event (by assumption, presumption, assertion, or evidence) is already known to have occurred.

$$P(A \mid B) = \frac{P(A, B)}{P(B)}$$

Stats Background: Conditional Probability

What is P(A)?

$P(A) = 0.30 + 0.10 + 0.12 = 0.52$

What is $P(A|B_1)$?

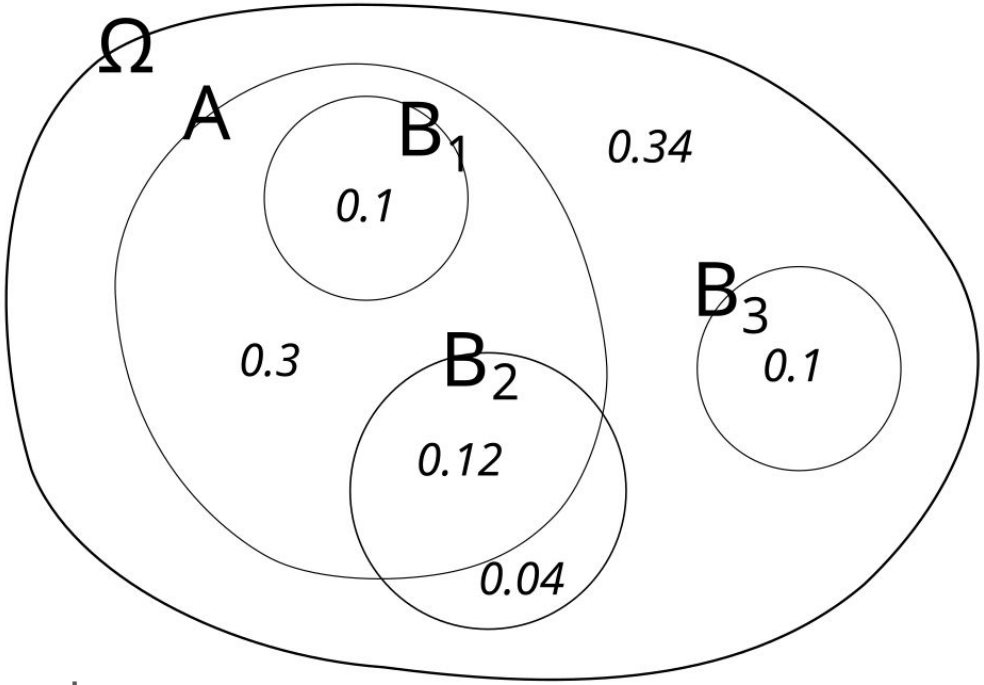
$P(A|B_1) = 1$

What is $P(A|B_2)$?

$P(A|B_2) = 0.12 \div (0.12 + 0.04) = 0.75$

What is $P(A|B_3)$?

$P(A|B_3) = 0$



Reduce the search space

Stats Background: Bayes Theorem

“reverse the conditional”

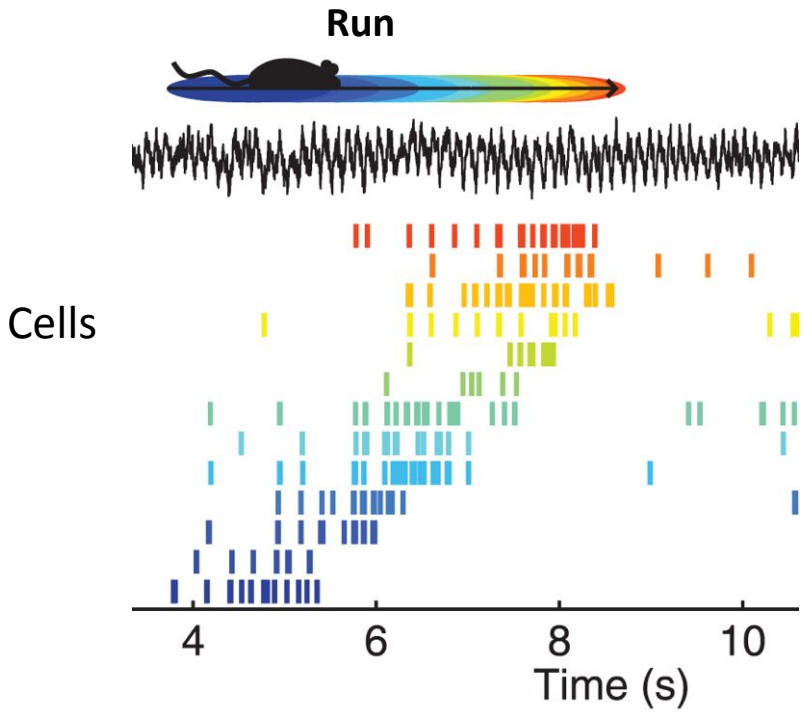
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Stats Background: Conditional Independence

Conditional independence means that once you know C, learning B tells you nothing more about A

$$P(A, B \mid C) = P(A \mid B, C)\, P(B \mid C) = P(A \mid C)\, P(B \mid C)$$

Bayesian Decoder



Want to infer:

$$p\left(x_t \middle| \Delta N_{1:t}^{1:C}\right)$$

Probability of latent position x given spiking from C cells at time t

t	Time
$\Delta N^{1:C}$	Observed spikes from C cells
x	Latent position

Adapted from Diba & Buszaki 2007

What is a latent state?

A latent state is **not directly observable,** but must be inferred via mathematical model through other observations.

In this case, the latent state variable **x is the mental position of the animal.**

Our observations are the **spike times.**

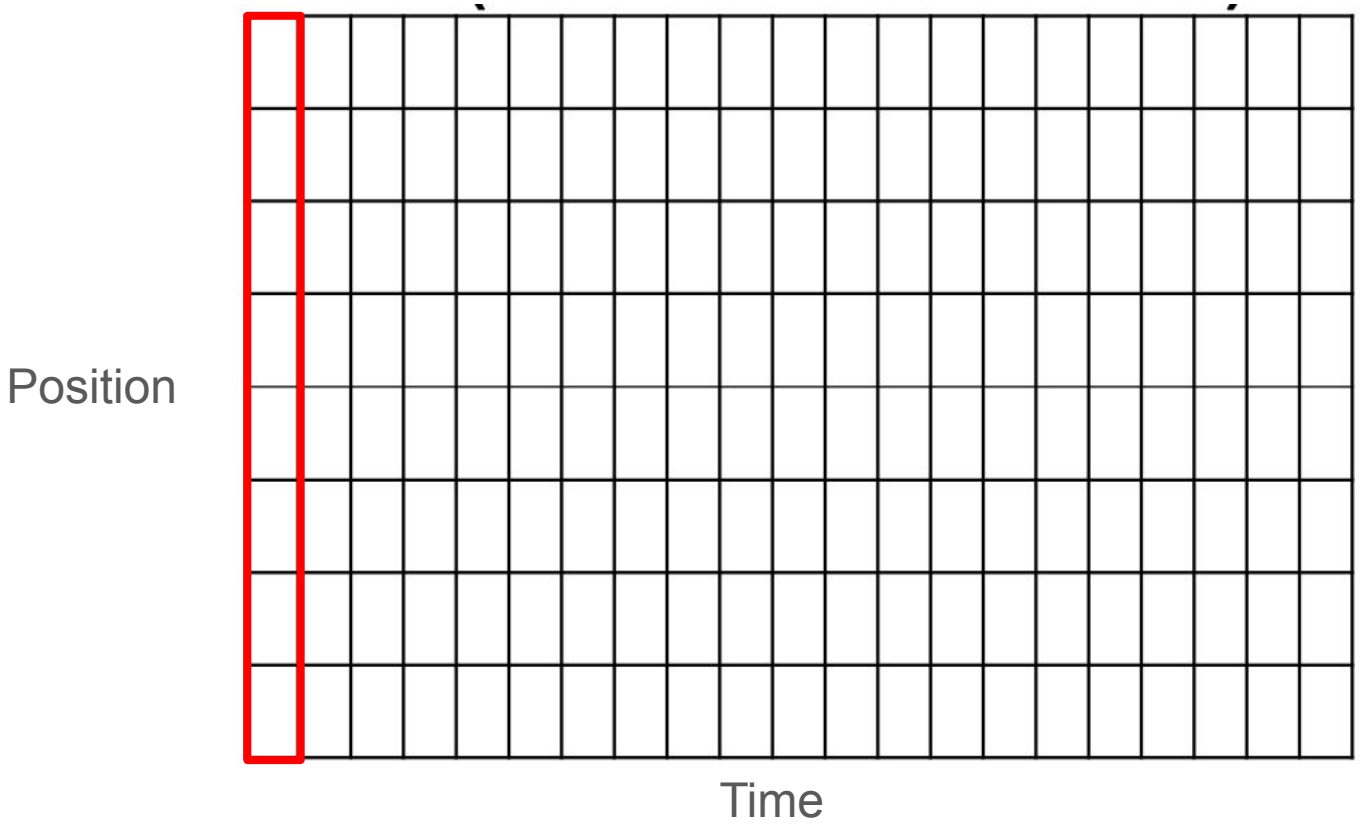
The **actual position of the animal** is observable.

Want to infer:

$$p\left(x_t \middle| \Delta N_{1:t}^{1:C}\right)$$

Probability of latent position x given spiking from C cells at time t

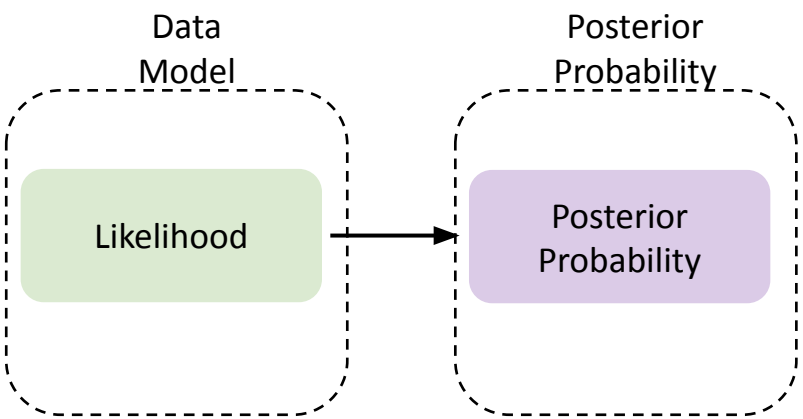
Bayesian Decoder



Subdivide time and position into bins

We will consider each time bin

Bayesian Decoder: for each time bin



$$p(x_t|\Delta N_{1:t}^{1:C}) \propto p(\Delta N_t^{1:C}|x_t) * p(x_t)$$

Probability of Position

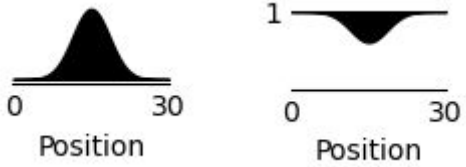
Poisson Likelihood

Prior Probability of Position

If spike

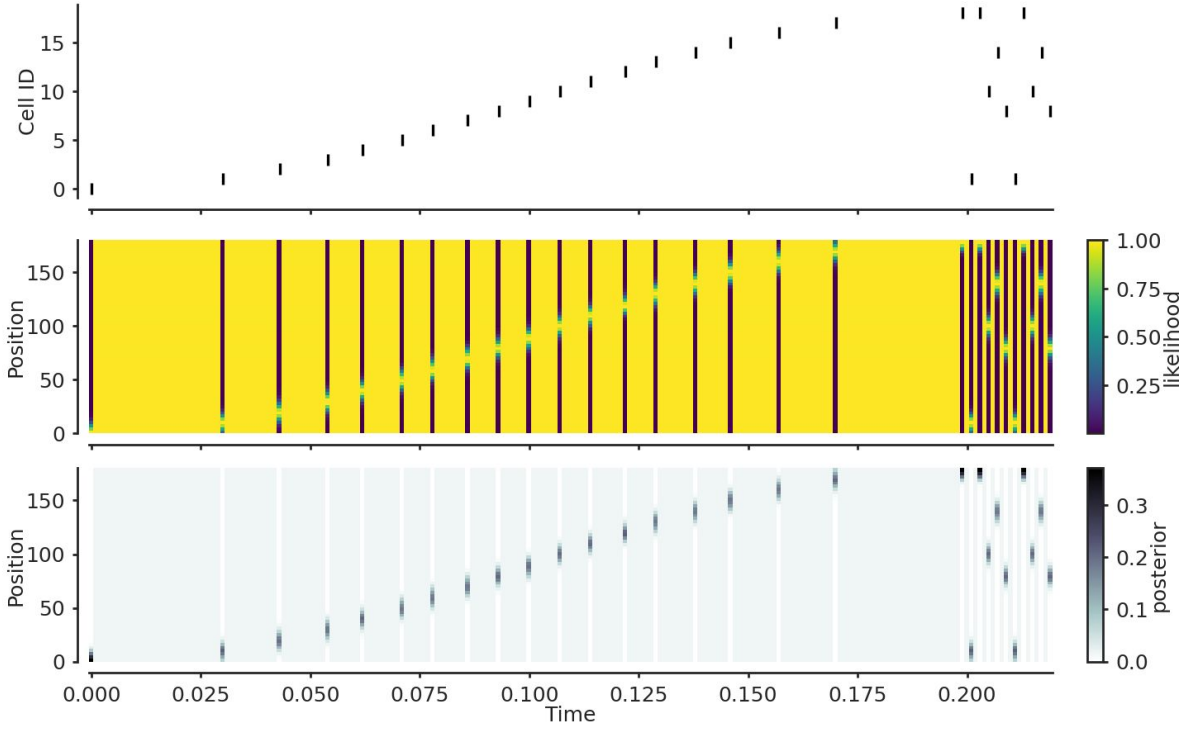
$$\prod_{i=1}^{C} [\lambda_i(t_k \mid x_k)\Delta_k]^{N_{t_k}^i} * e^{-\lambda_i(t_k \mid x_k)\Delta_k}$$

If no spike



t	Time
$\Delta N^{1:C}$	Observed spikes from C cells
x	Latent position

Example on simulated data



Likelihood

Posterior Probability

Assumptions and issues with the Bayesian decoder

- Numerical stability of likelihoods
- Choice of bin size
- Choice of smoothing in the encoding model rate maps
- Assumes we have the correct encoding model
- Assumes we have spike sorted correctly
- Assumes we have the correct decoding model
- Assumes we have conditional independence of neurons
- Computational efficiency
- Likelihood uses the expected (average) rate
- High-rate neurons contribute more to the likelihood; informativeness depends on tuning, not just mean rate
- Behavioral sampling bias / occupancy bias
- No-spike likelihood and incomplete field coverage (cell-sampling bias)
- Zero rates
- Choice of prior

Today…

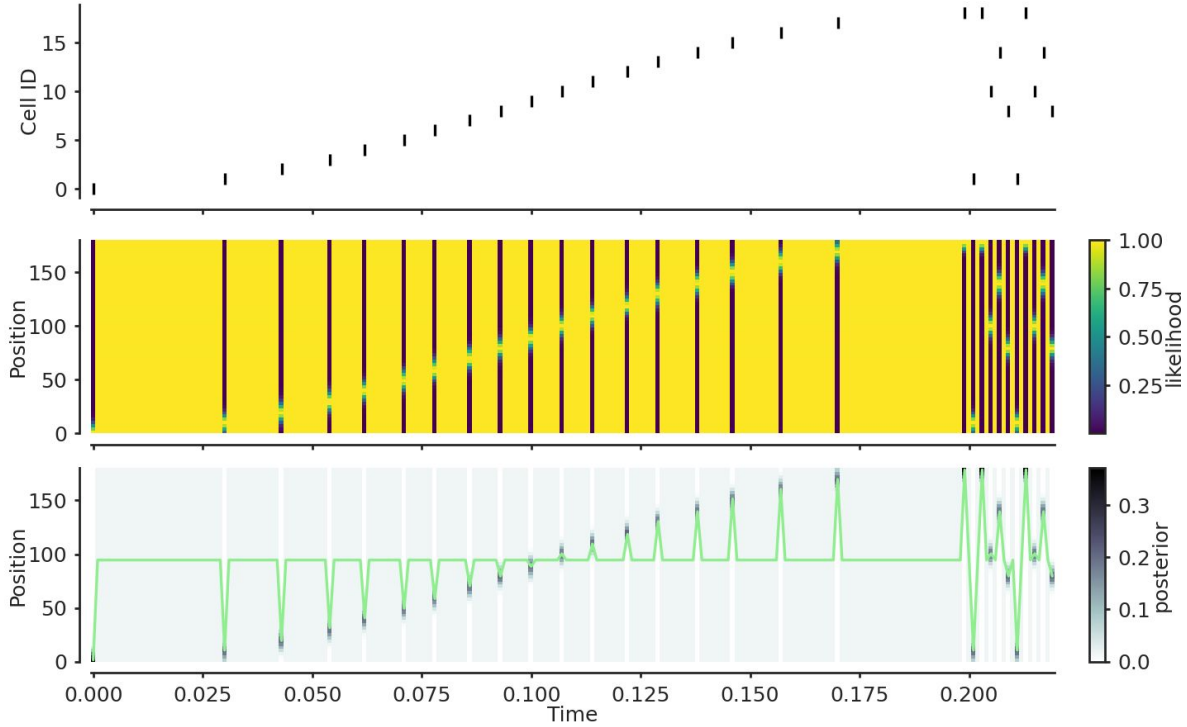
How do we estimate better trajectories?

How do we avoid spike sorting?

How do we classify different types of trajectories?

How can we think about uncertainty for these models?

Maximum A Posteriori Estimate (MAP) finds the maximum probable position in each time bin.

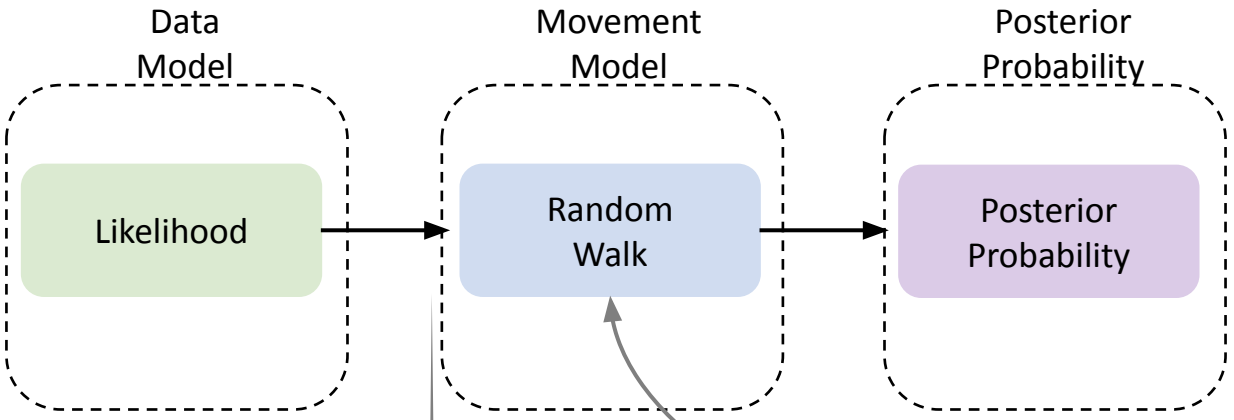


Likelihood

Posterior Probability

Most Probable Position

State space model: Data Model + Movement Model



Prior on Movement Dynamics

$$p(x_t|\Delta N_{1:t}^{1:C}) \propto p(\Delta N_t^{1:C}|x_t) \propto p(\Delta N_t^{1:C}|x_t) * \int p(\Delta N_t^{1:C}|x_{t-1}) * p(x_{t-1}|\Delta N_{1:t-1}^{1:C})\, dx_{t-1}$$

Current Probability Of Position

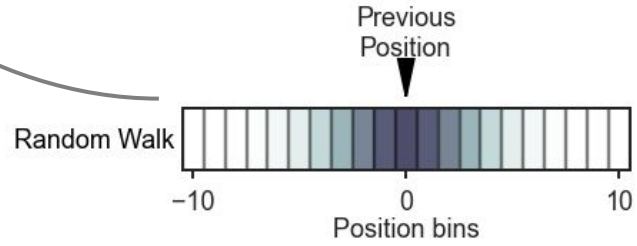
Probability of Position

Poisson Likelihood

Movement Model

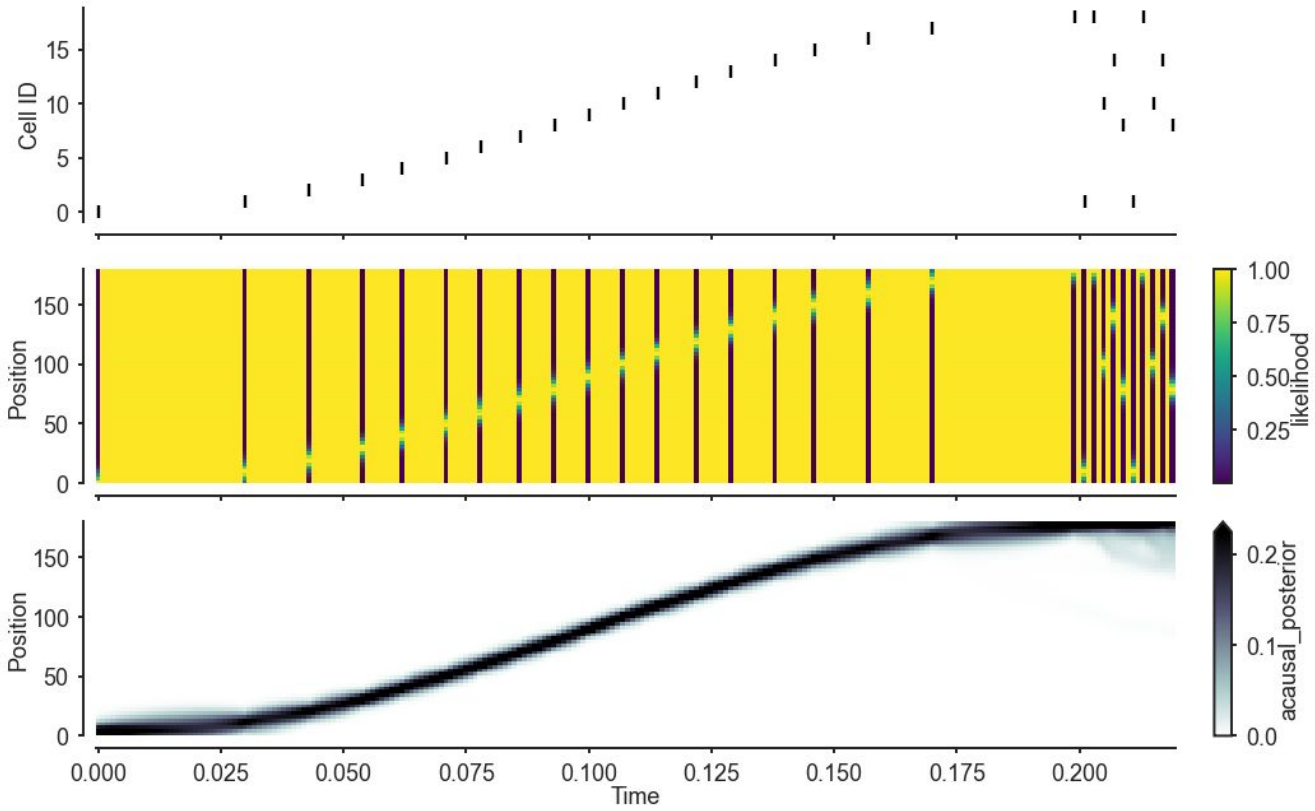
Prior

Previous Probability of Position



Zhang…Sejnowski (1998)
Brown, Frank…Wilson (1998)

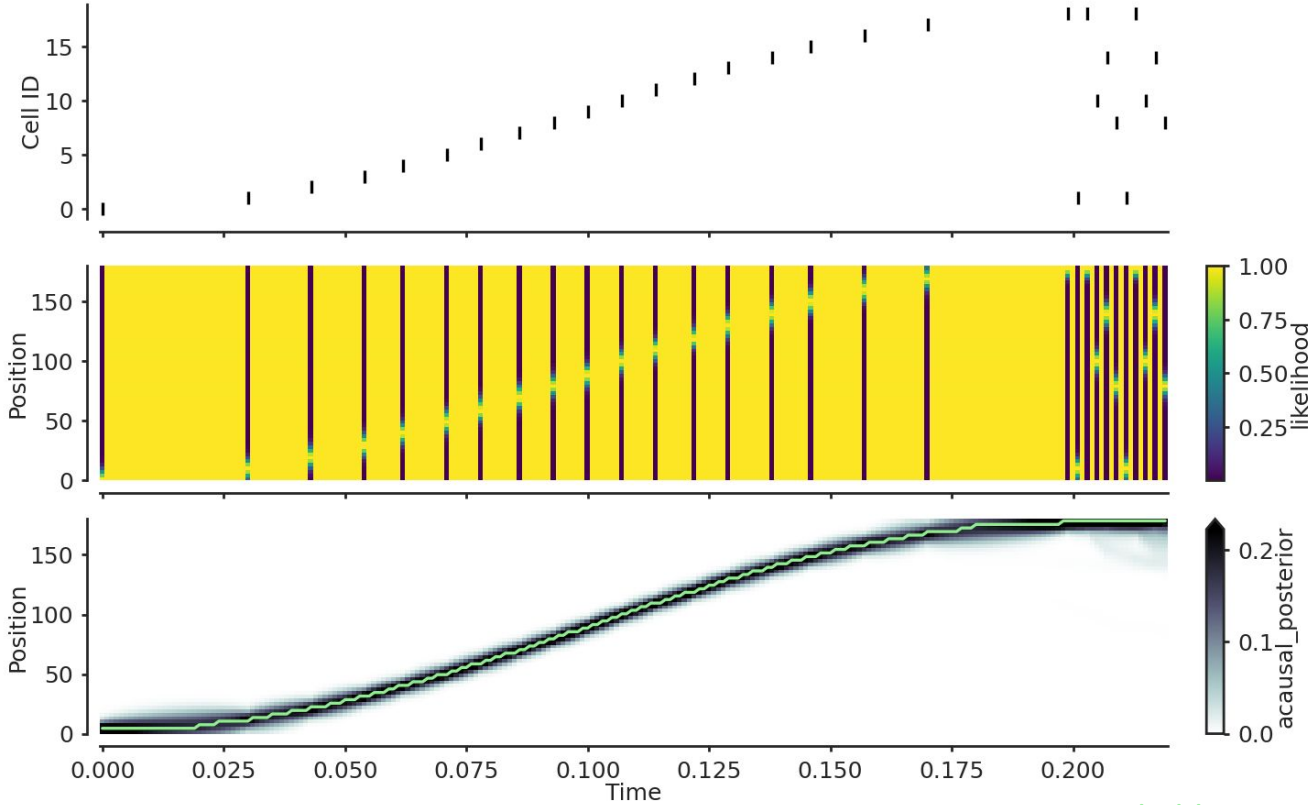
State space model captures continuous sequences



Likelihood

Posterior Probability

Most probable position is smoother than before



Likelihood

Posterior Probability

Most Probable Position

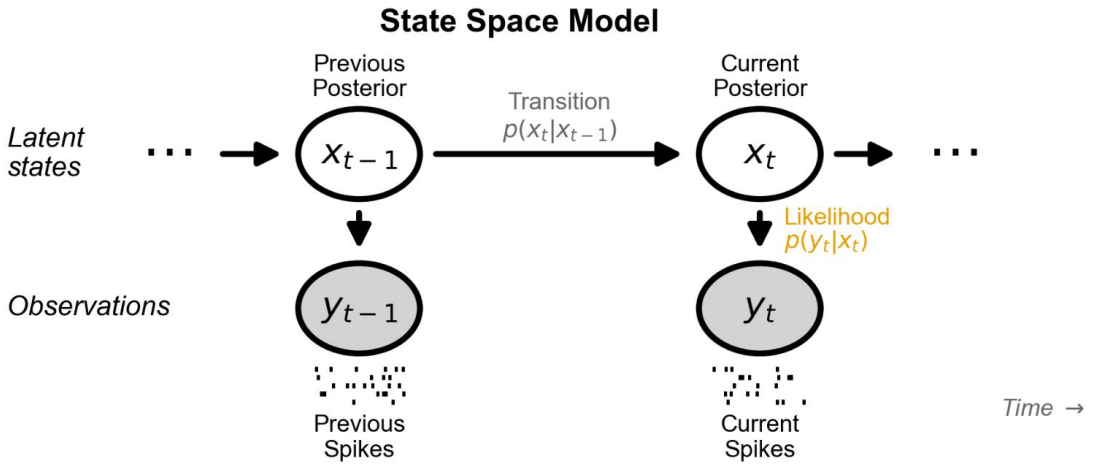
Different types of state space models you may have heard of

Hidden markov model (HMM) - discrete latent states

Kalman filter/smoother - continuous latent states, uses Gaussians to approximate posterior

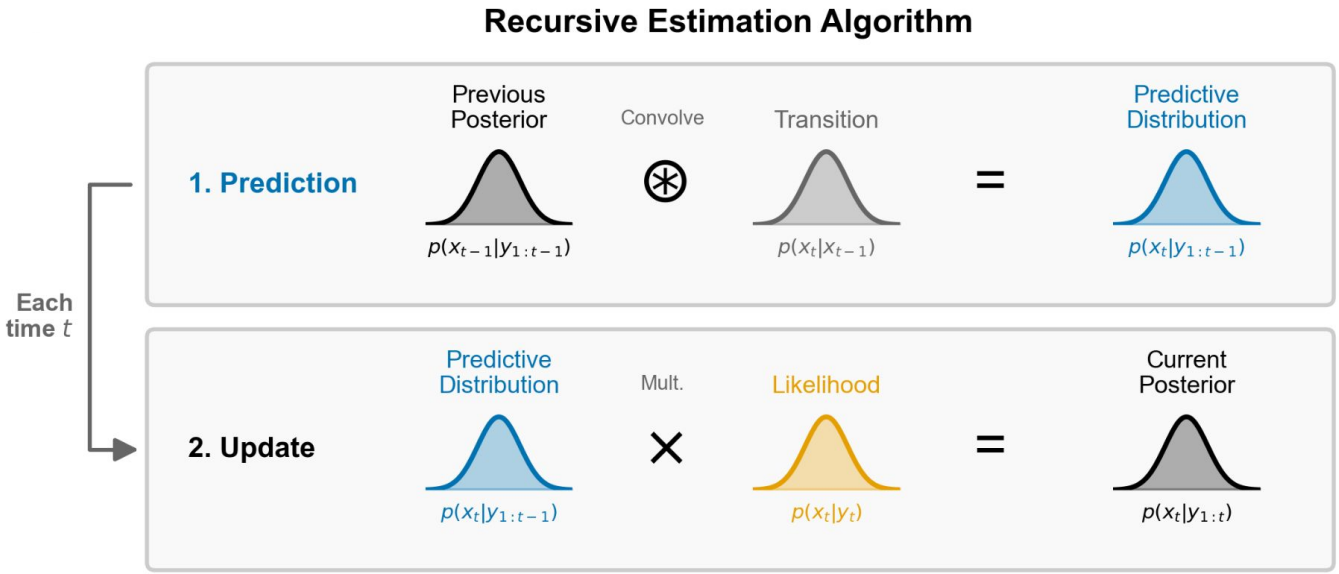
What we are doing here is in-between; we are approximating a continuous latent states with discrete states.

Graphical model representation of a state space model



Arrows indicate conditional dependence

Recursive estimation



State space models may be updated efficiently (causally)

For non-real time analyses, you want to update acausally (Rauch–Tung–Striebel smoother)

This means you estimate the filter forwards in time (causally), and then run it backwards in time (like filtfilt).

Predictive Transition Filter

$$p(\mathbf{x}_{k+1} \mid \mathbf{y}_{1:k}) = \int p(\mathbf{x}_{k+1} \mid \mathbf{x}_k)\, p(\mathbf{x}_k \mid \mathbf{y}_{1:k})\, \mathrm{d}\mathbf{x}_k,$$

Future Smoother

$$p(\mathbf{x}_k \mid \mathbf{y}_{1:T}) = p(\mathbf{x}_k \mid \mathbf{y}_{1:k}) \int \left[\frac{p(\mathbf{x}_{k+1} \mid \mathbf{x}_k)\, p(\mathbf{x}_{k+1} \mid \mathbf{y}_{1:T})}{p(\mathbf{x}_{k+1} \mid \mathbf{y}_{1:k})} \right] \mathrm{d}\mathbf{x}_{k+1}$$

Current Smoother

1. Predict forward from the filter
2. Reweight the filter using the smoother-vs-filter mismatch at k+1

Smoother intuition

Goal: estimate the latent state using *all* observations

Uses past + future data + the transition model

- Past gives evidence for where you were
- Future constrains where you must have been, via the transition model

Prefers states that can realistically lead to the future trajectory

- Upweight latent state values that transition into likely future states
- Downweight latent state values that would require implausible jumps

Algorithm view: filter forward, then pass information backward through the dynamics

- Forward pass: causal “online” belief
- Backward pass: uses future evidence to refine earlier states

Ideal-observer perspective

- “Given everything before *and after* time t, what state was most likely at t?”

Why smoothing helps vs filtering

- Filtering is causal: it can take time (several bins) to accumulate enough evidence
- Smoothing can attribute that evidence to the *earlier* time where the state change actually occurred

Why not use sliding windows?

Sliding windows borrow evidence "by distance in time."

- They mix spikes from many nearby bins (often including far-away bins) with fixed weights.

They implicitly assume the state is constant within the window.

- Windowing stabilizes estimates by *averaging observations*, which blurs transitions and can introduce lag (or acausality if centered).

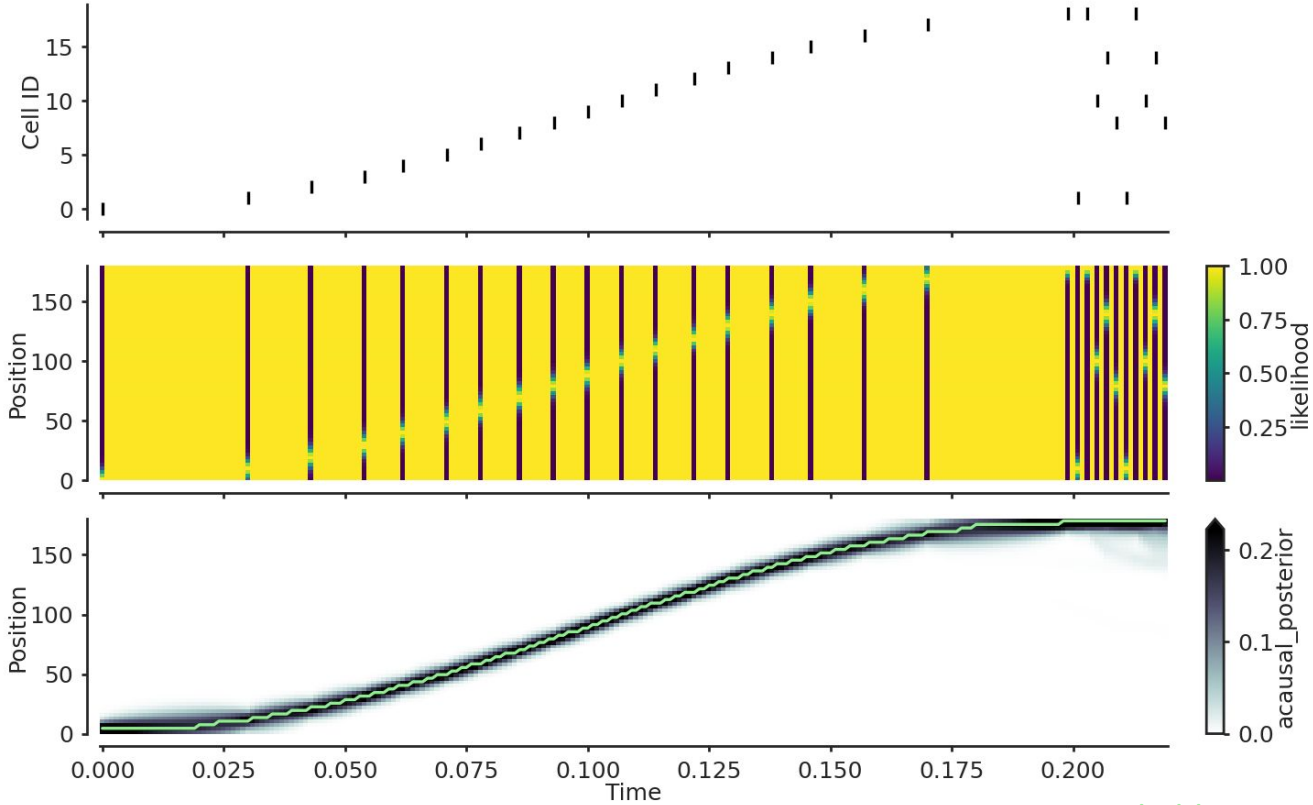
State-space models borrow evidence "by plausibility."

- With the Markov assumption, the state only connects to adjacent steps (transition from previous or future time step)
- You *can* extend this beyond one step, but the key is that dependence is mediated through dynamics, not an arbitrary window.

The transition matrix is an explicit claim about how the latent state changes.

- It pools information forward/backward **only in ways consistent with realistic movement** (e.g., small jumps more likely than large jumps).

There seem to be two different “modes” in the data, but we only captured one



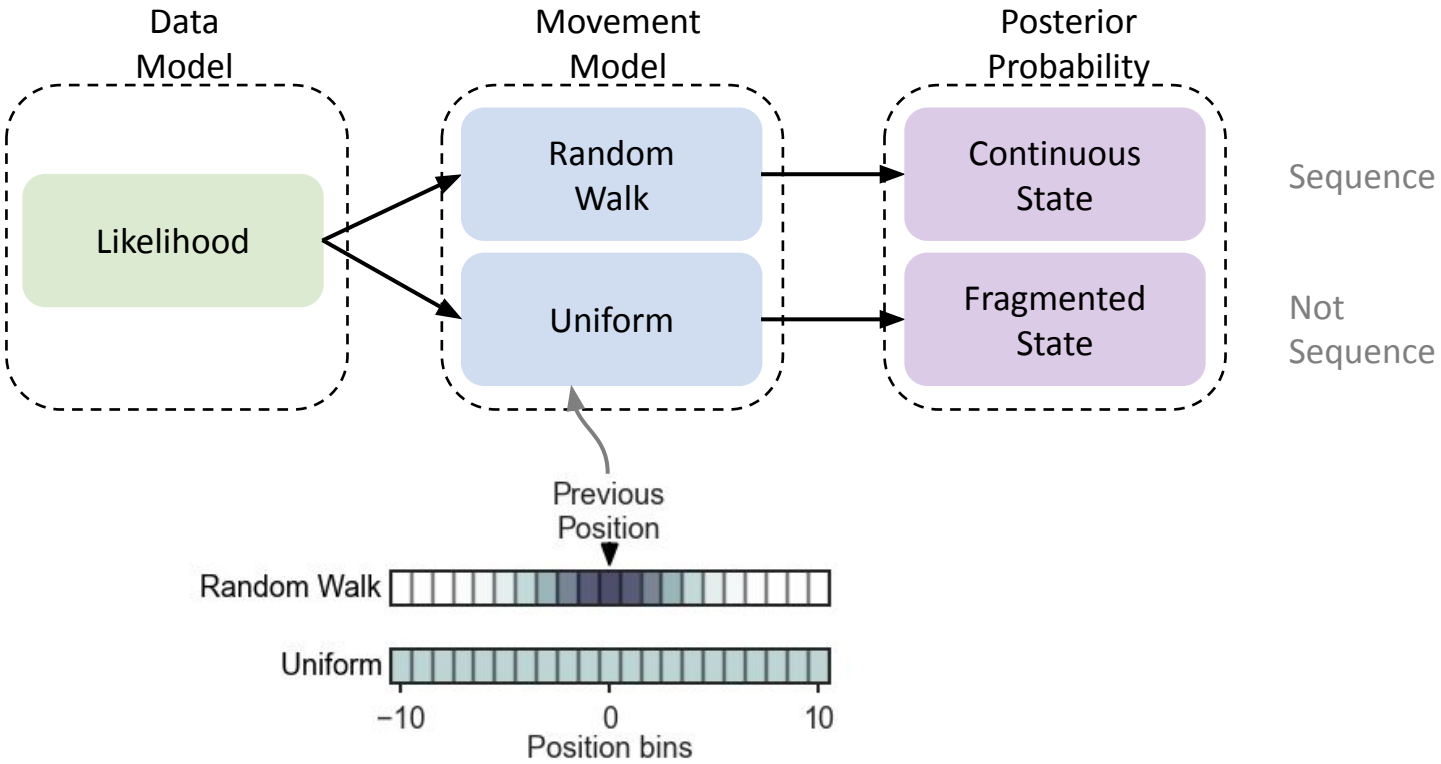
Likelihood

Posterior Probability

Most Probable Position

How might we deal with this?

What if we could allow multiple movement (transition) models?



Associate each movement model (continuous state) with a discrete state and allow switching between them

Continuous

$$p(x_t|\Delta N_{1:t}^{1:C}) \propto p(\Delta N_t^{1:C}|x_t) * \int p(x_t|x_{t-1}) * p(x_{t-1}|\Delta N_{1:t-1}^{1:C})\, dx_{t-1}$$

Current Probability of Position | Poisson Likelihood | Movement Model | Previous Probability of Position

Discrete and Continuous

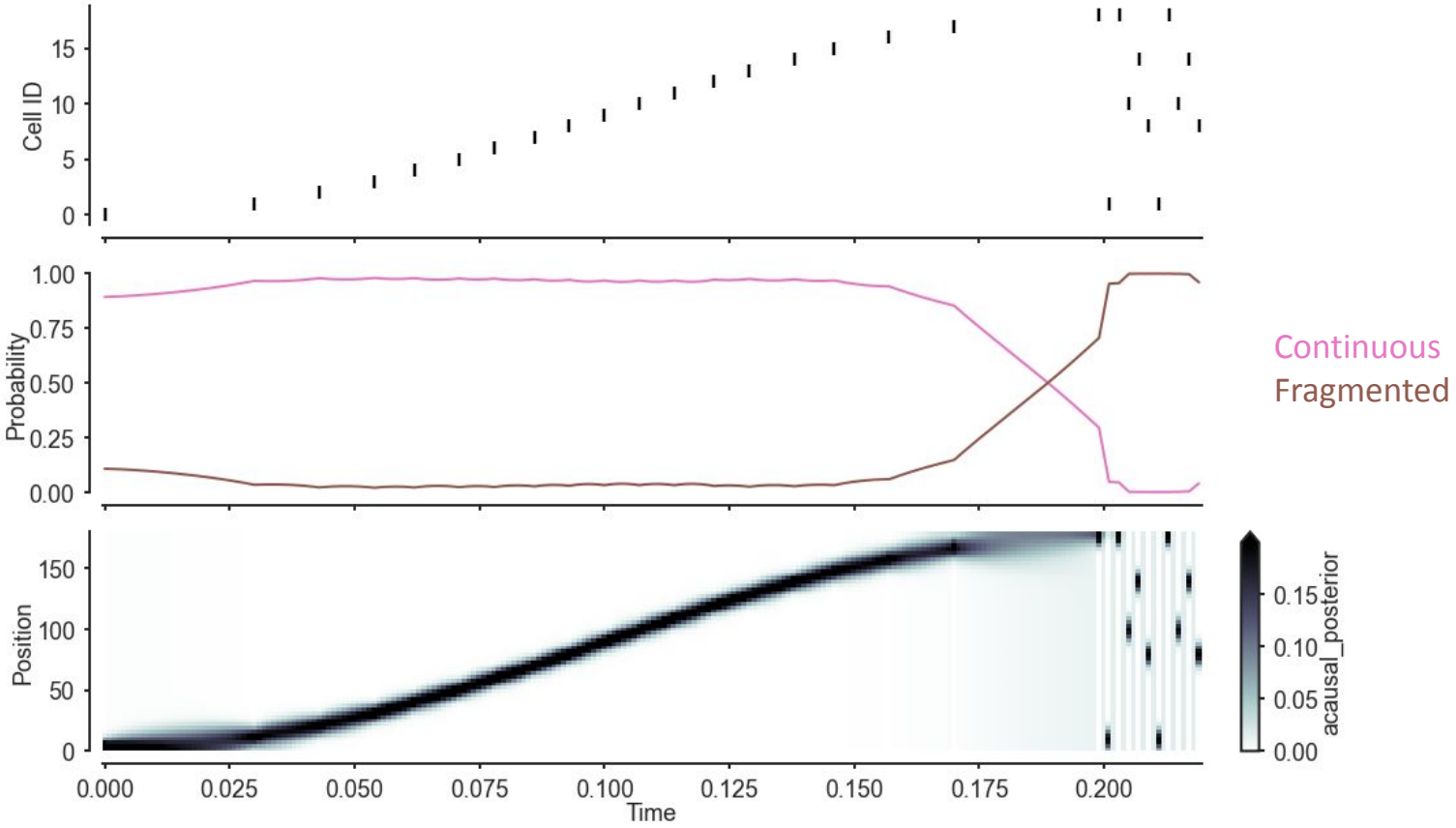
$$p(x_t, I_t|\Delta N_{1:t}^{1:C}) \propto p(\Delta N_t^{1:C}|x_t, I_t) * \sum_{I_{t-1}} \int p(x_t|x_{t-1}, I_t, I_{t-1}) * Pr(I_t|I_{t-1}) * p(x_{t-1}, I_{t-1}|\Delta N_{1:t-1}^{1:C})\, dx_{t-1}$$

Current Probability of Position | Poisson Likelihood | Movement Models | Discrete State Transition | Previous Probability of Position

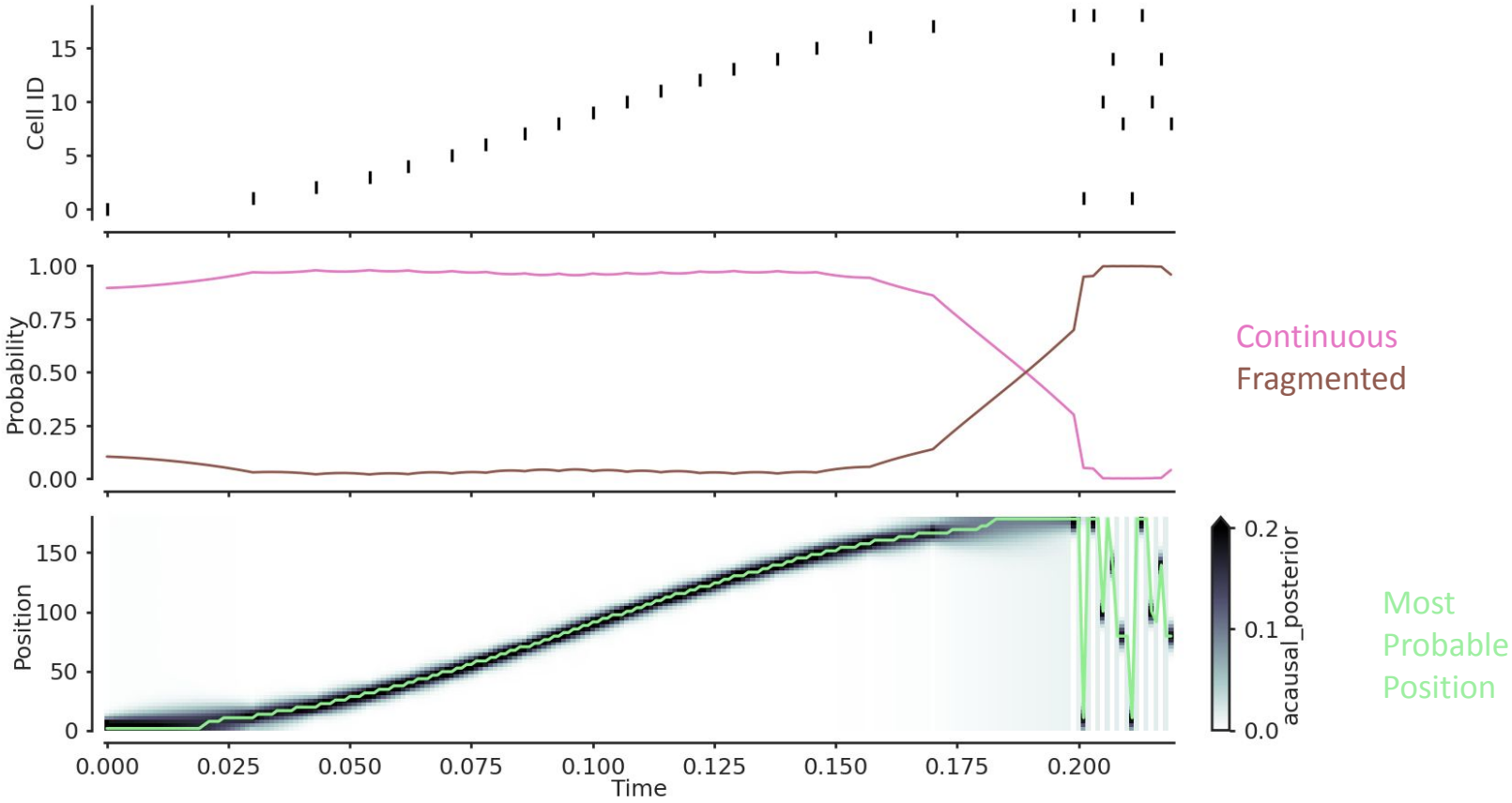
$$I_t = \begin{cases} \text{Continuous} \\ \text{Fragmented} \end{cases}$$

t	Time
$\Delta N^{1:C}$	Observed spikes from C cells
x	Latent position
I	Latent discrete state

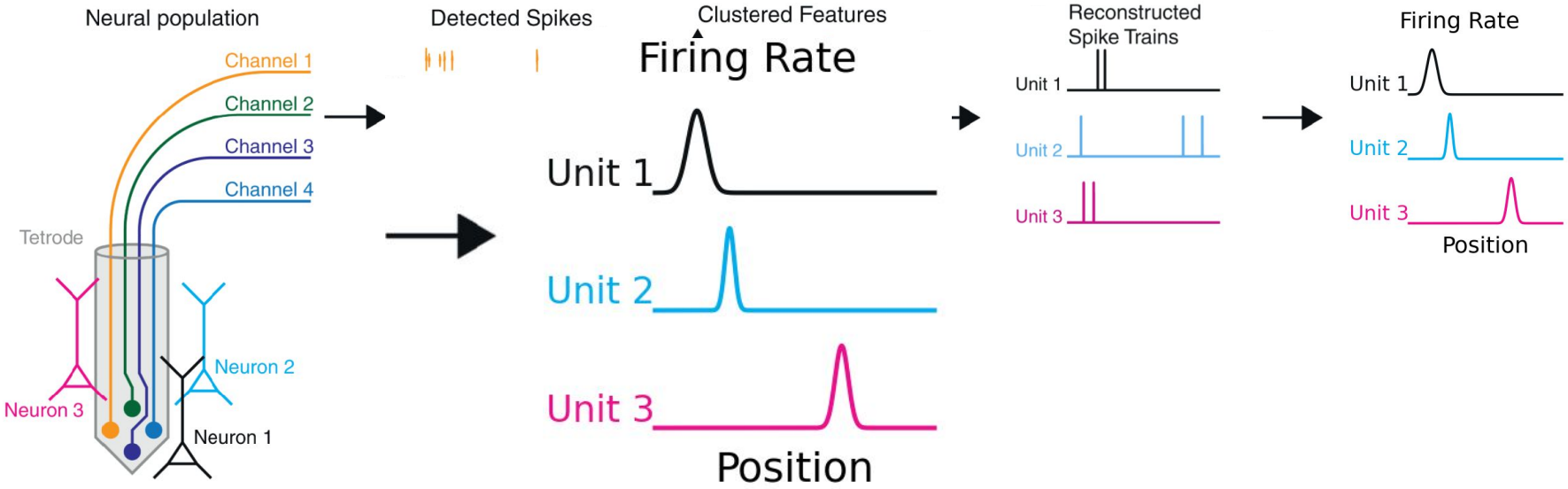
Switching state space model prevents misclassifying the end



Most probable position corresponds to our intuition

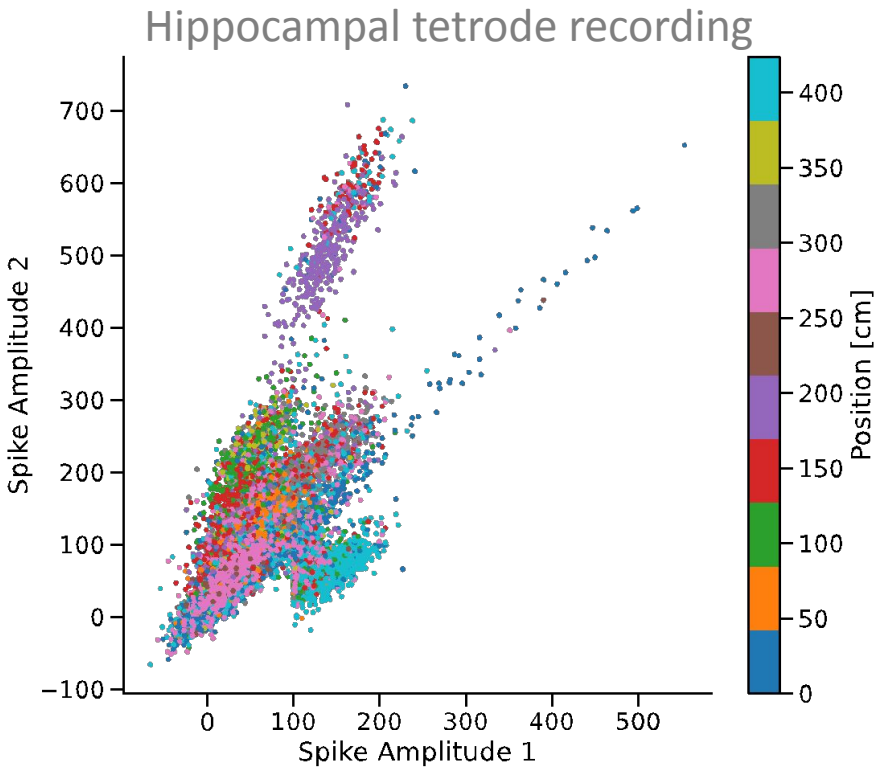


Getting the most out of the data by avoiding spike sorting



Adapted from:
Einevoll…Harris (2013)

Spike sorting loses information



Disadvantages:

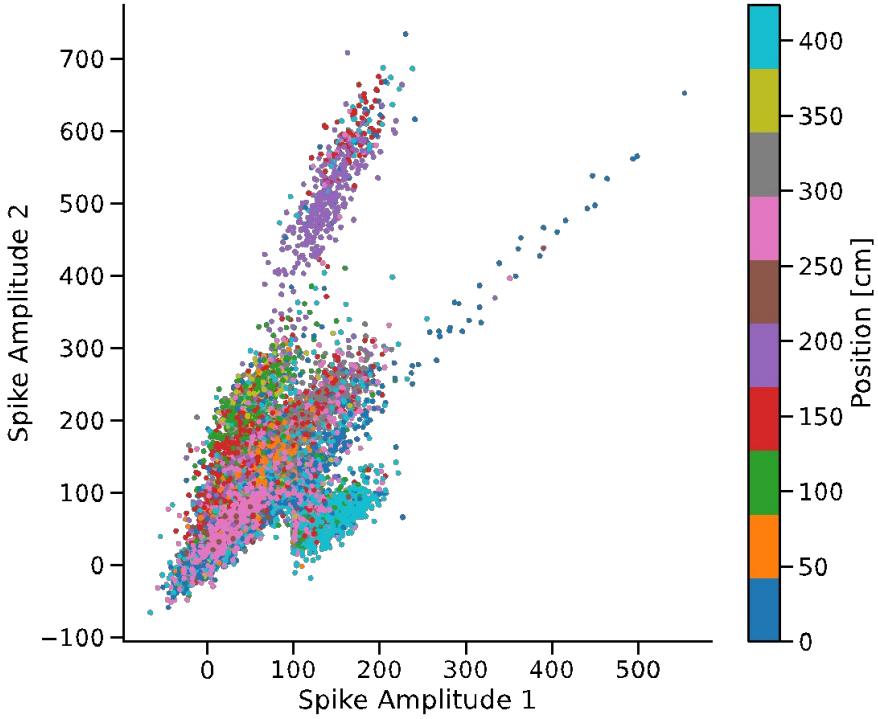
- Lose information from spikes not in cluster
- Experimenter variability

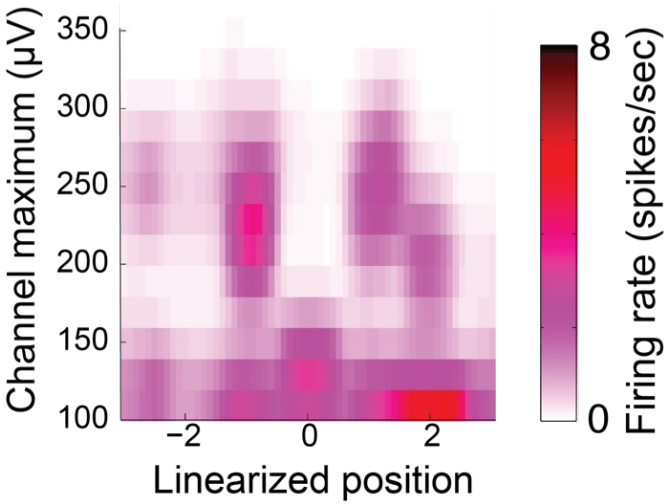
Cox and Isham (1980)
Chen…Wilson (2012)
Kloosterman…Chen, Wilson (2014)
Deng…Frank, Eden (2015)

Data from Dr. Abhilasha Joshi

Clusterless: Relate waveform features to position by rate (marked point process)

Hippocampal tetrode recording





$$p\left(\Delta N_t^{1:E}, \{\vec{m}_{t,j}^{i}\}_{1:\Delta N_t^i}^{i=1:E} \middle| x_t, I_t\right)$$

Marked Point Process Likelihood

$\Delta N^{1:C}$ Observed spikes from C cells

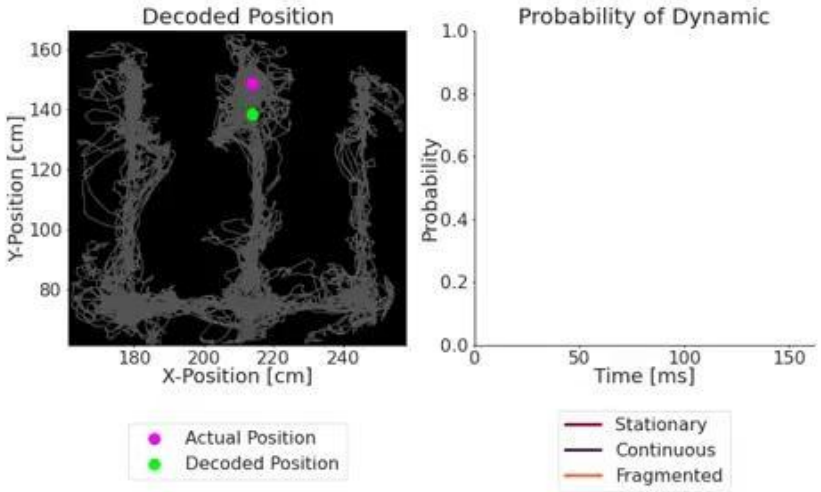
$\Delta N_t^{1:E}$ Observed multiunit spikes from E electrodes

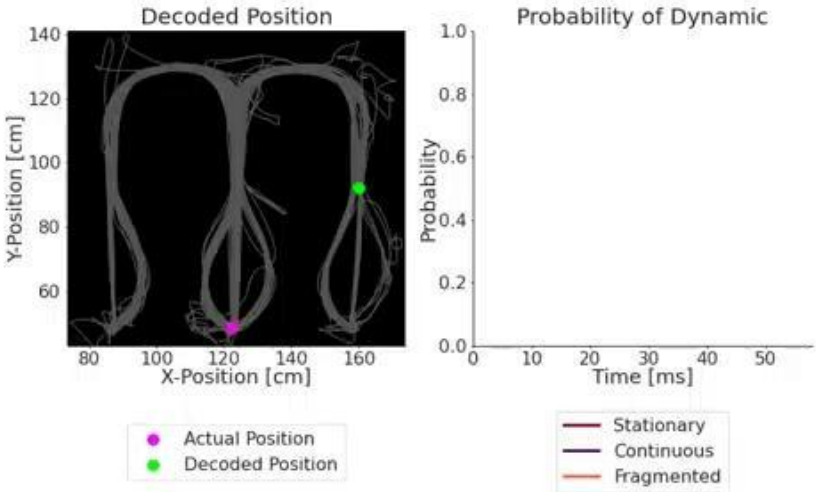
$\{\vec{m}_{t,j}^{i}\}$ Spike features

Cox and Isham (1980)
Chen…Wilson (2012)
Kloosterman…Chen, Wilson (2014)
Deng…Frank, Eden (2015)

Data from Dr. Abhilasha Joshi

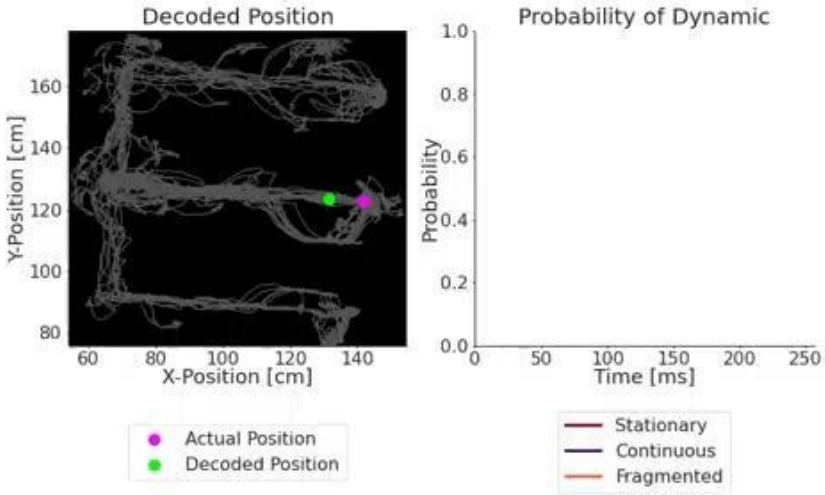
The state space model captures expected dynamics

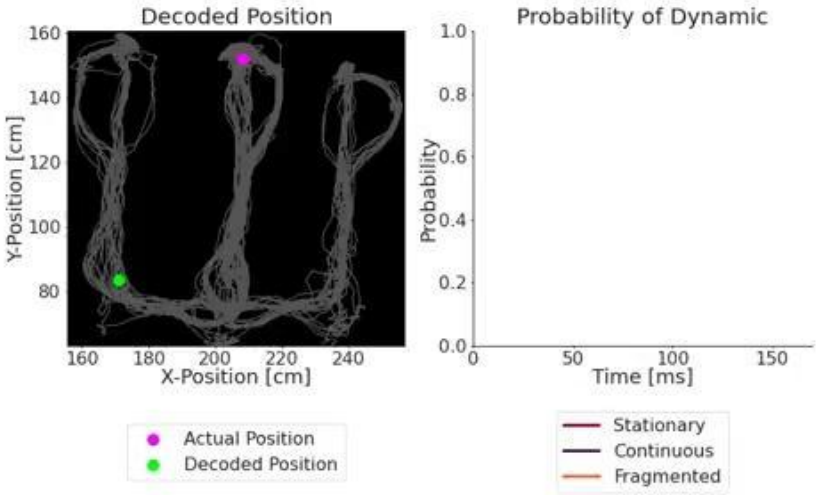




Denovellis…Frank *eLife* 2021

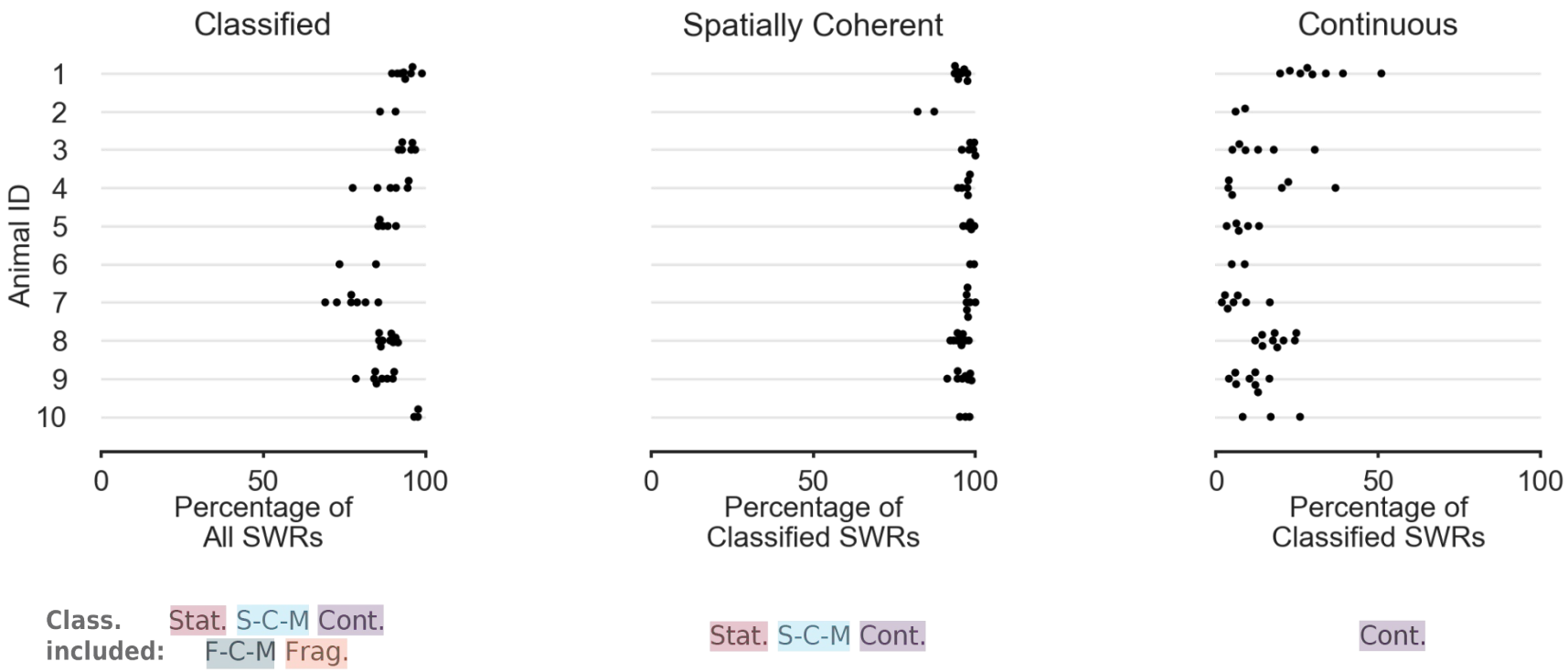
But also revealed a wider range of new dynamics





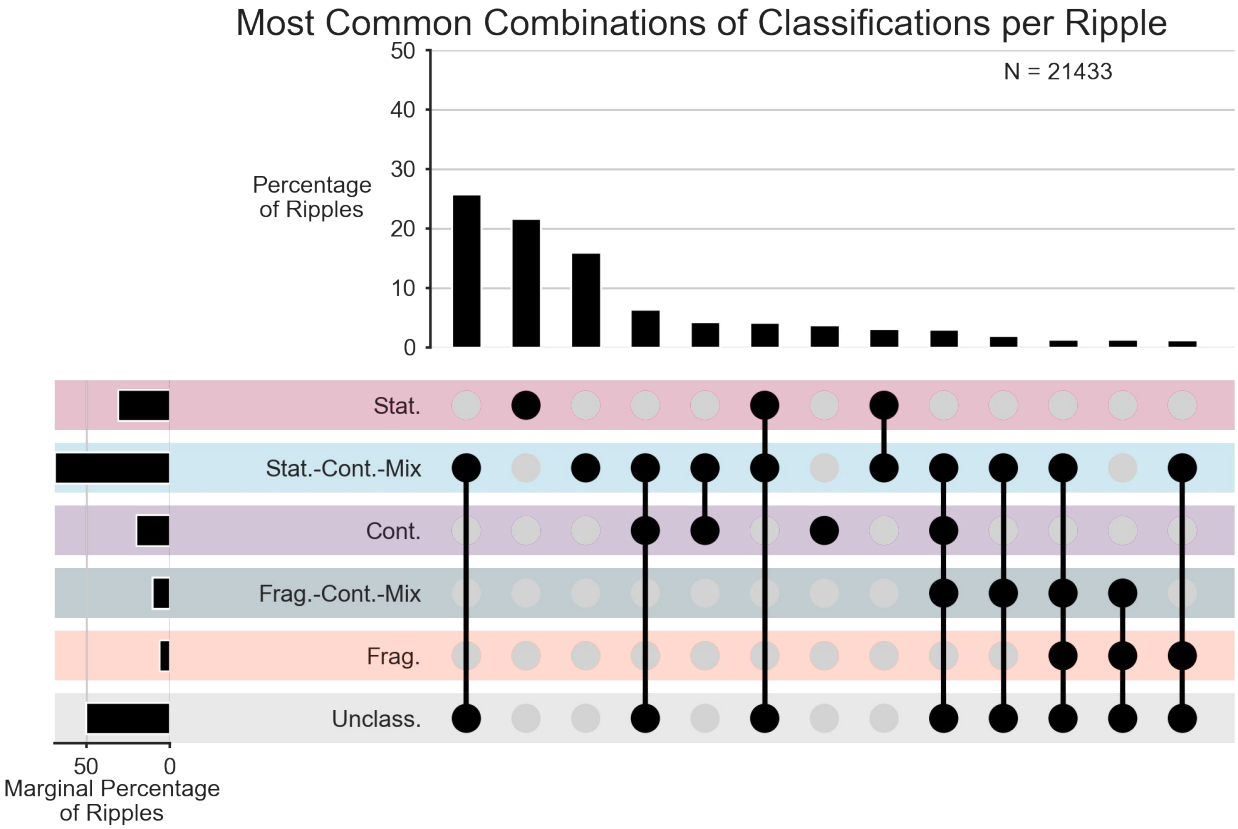
Denovellis…Frank *eLife* 2021

This model allowed us to characterize more SWRs than previous methods

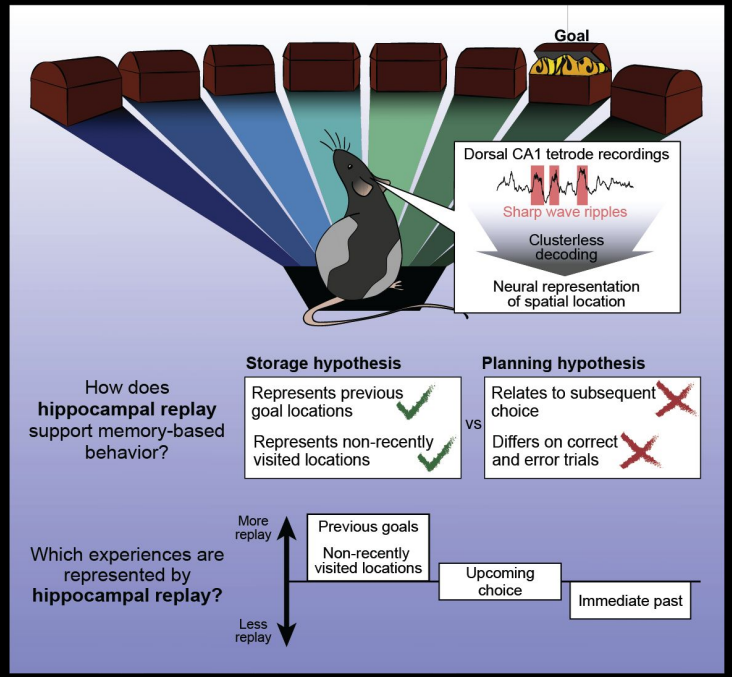


Denovellis…Frank *eLife* 2021

More slower replay events than previously known

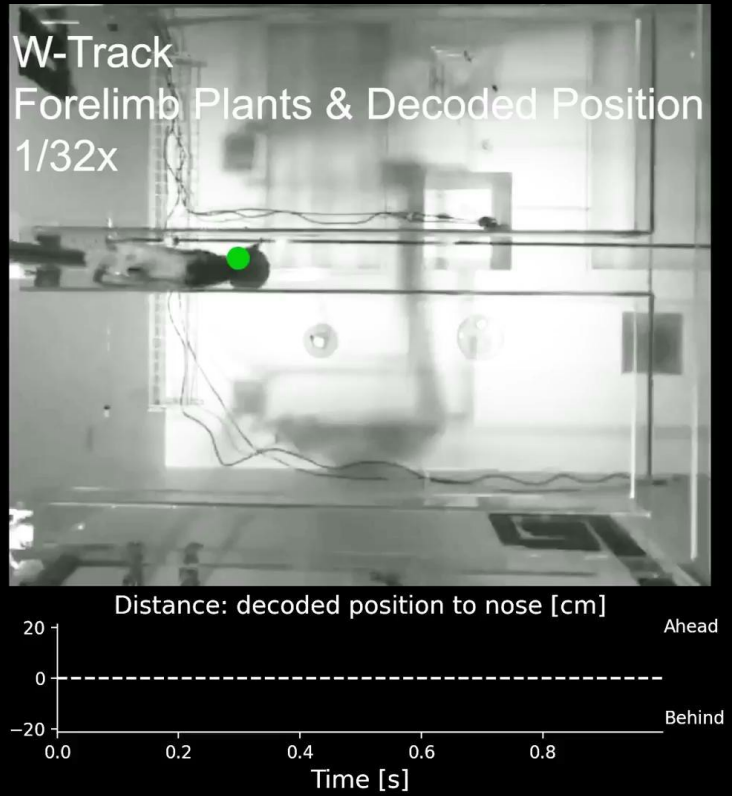


Denovellis…Frank *eLife* 2021



Replay reflects past experience

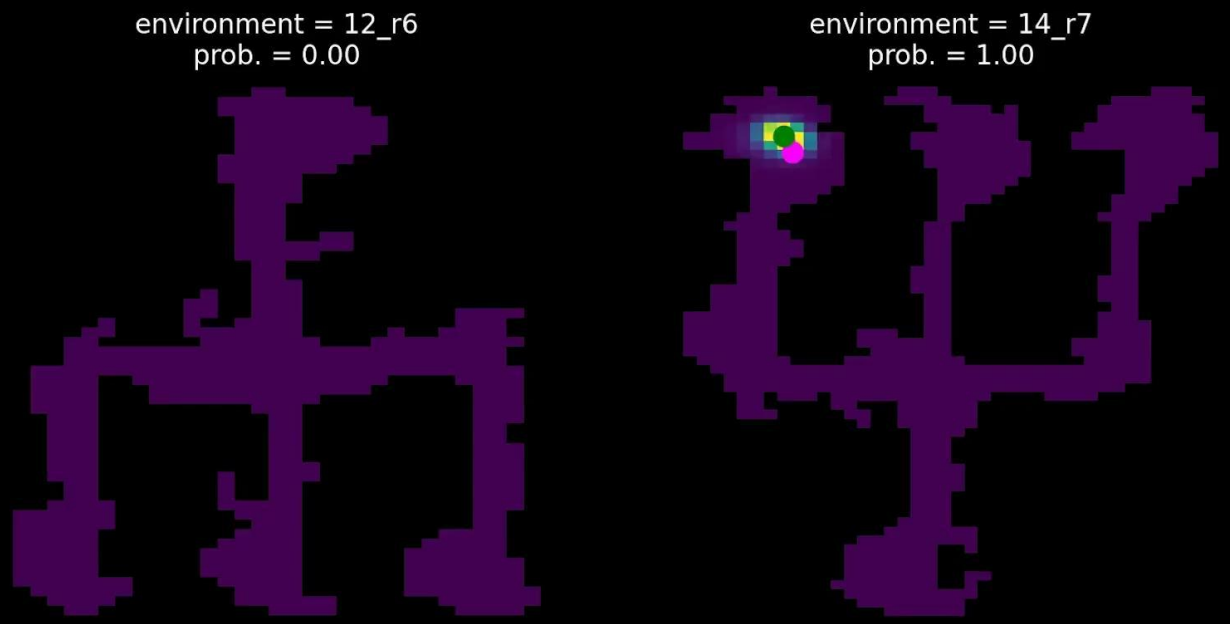
Gillespie…Denovellis…Frank *Neuron* (2021)



Theta sequences dynamically coordinate with stepping

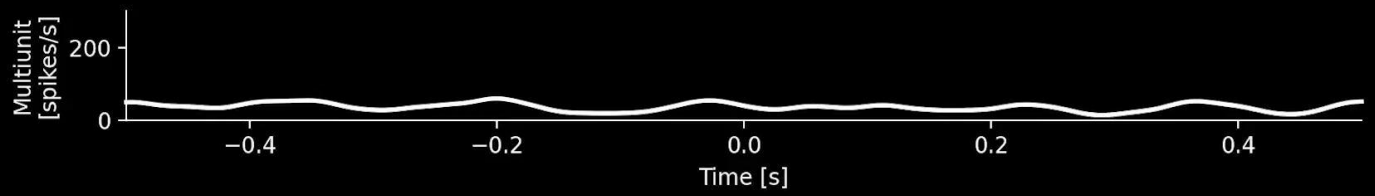
Joshi, Denovellis…Frank *Nature* (2023)

This framework extends beyond simple replay dynamics



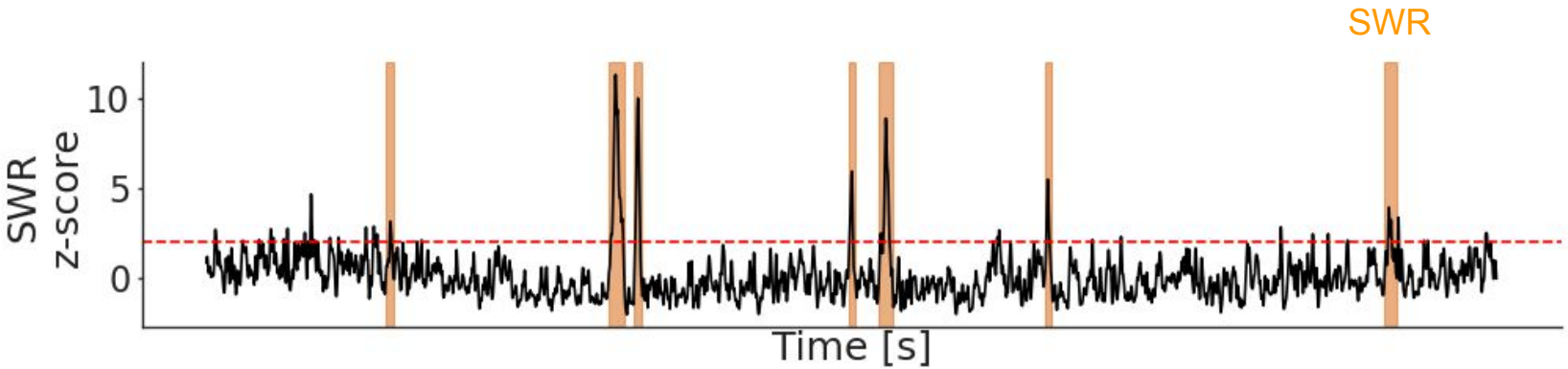
Animal's Position

Decoded Position



Data from Dr. Jennifer Guidera

3. When do replay sequences occur?



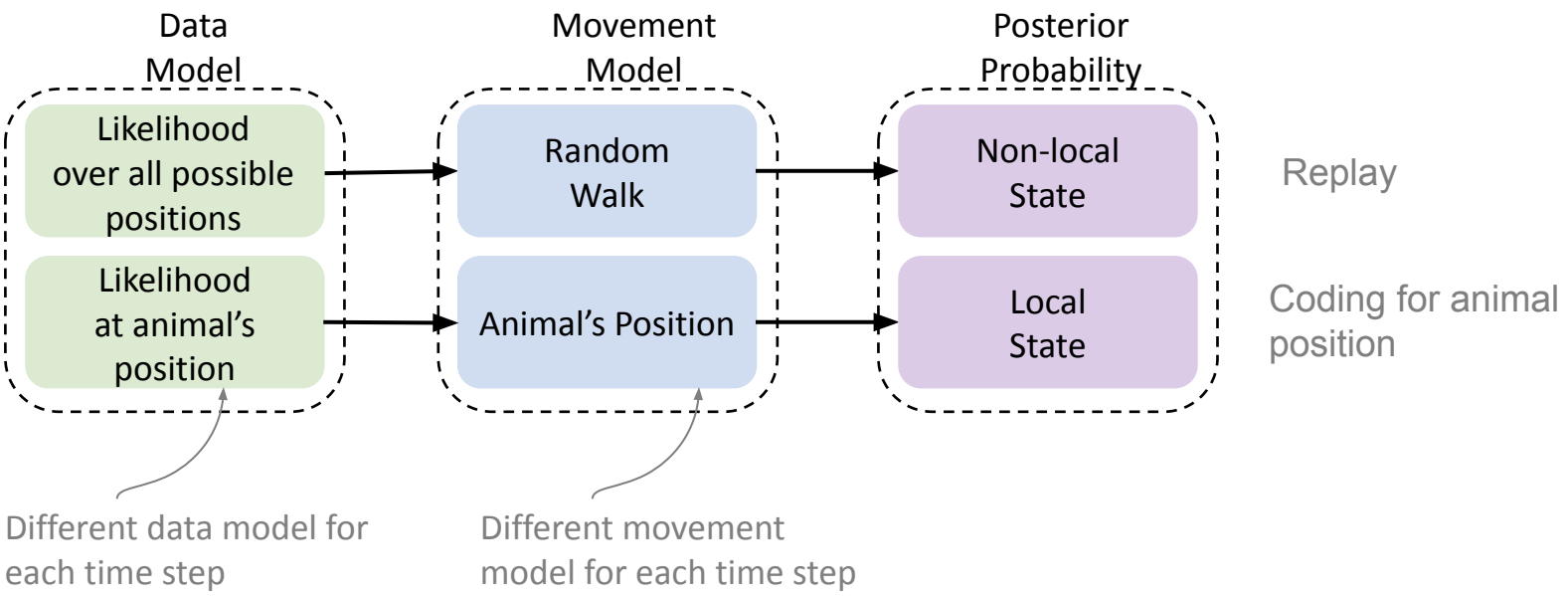
Typically **high ripple power** (150-250 Hz, SWR) is used to look for replay.

This is noisy, indirect, and has poor time resolution (>20 ms).

Can we use the representation to distinguish cognitive events from behaviorally-driven events?

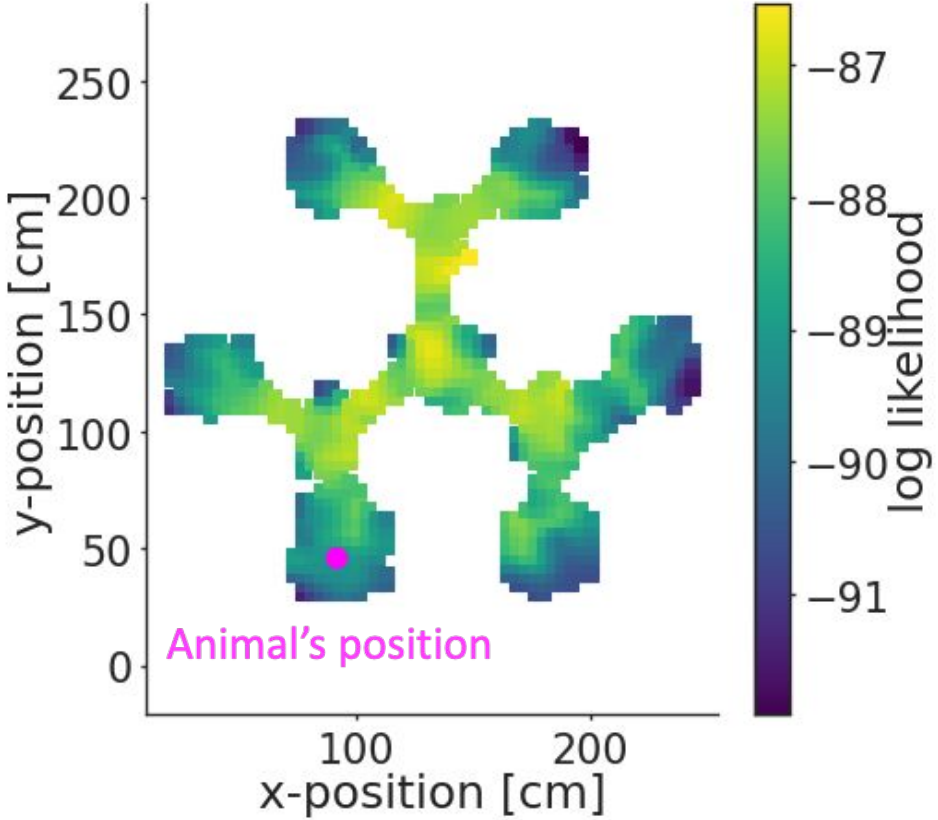
Data from Alison Comrie

Detecting non-local sequences: use the representation to distinguish cognitive events from behaviorally-driven events



Denovellis, Frank, Eden *IEEE* 2019

Likelihoods: Local vs. Non-Local at each time step



Non-Local Likelihood: every possible bin

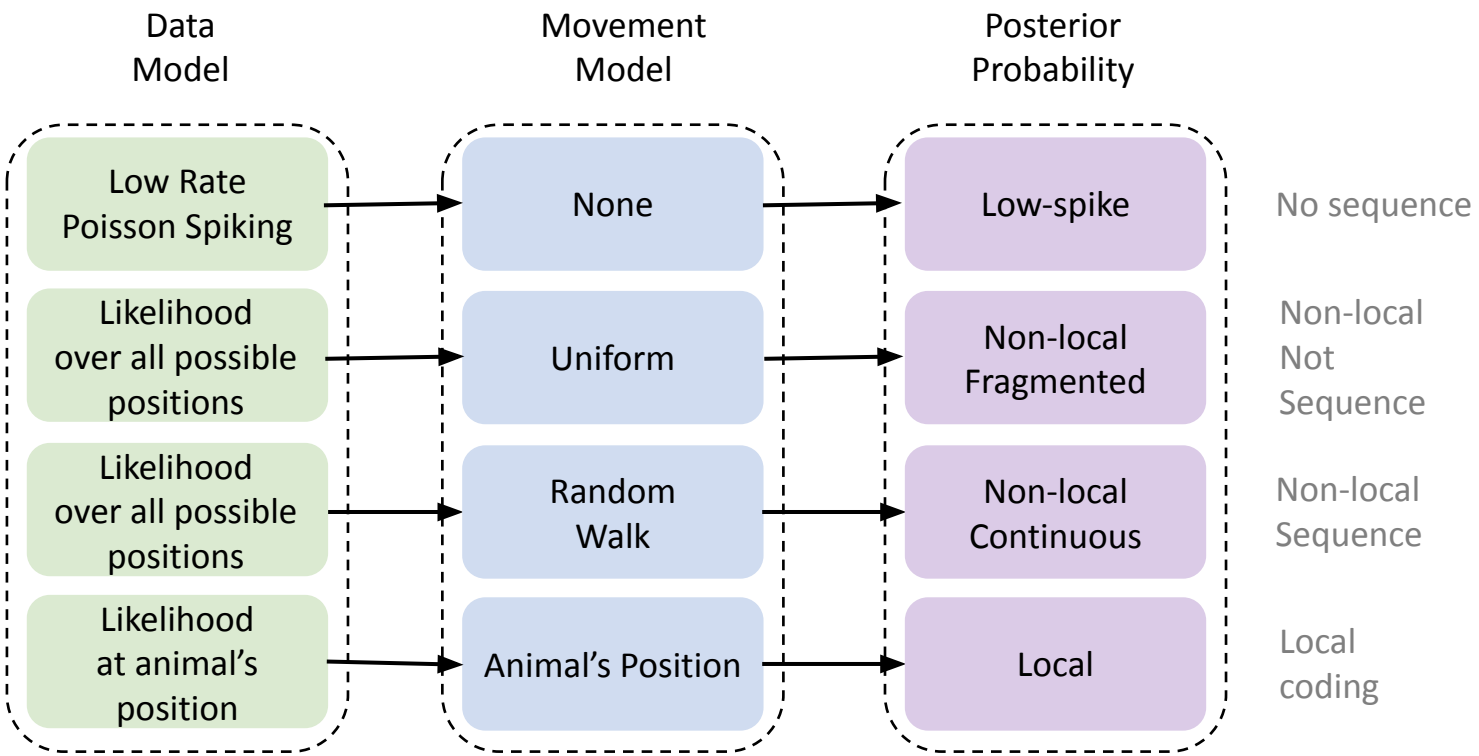
Local Likelihood: bin at animal's position

$$p\left(\Delta N_t^{1:E}, \{\vec{m}_{t,j}^{i}\}_{1:\Delta N_t^i}^{i=1:E} \middle| x_t, I_t\right)$$

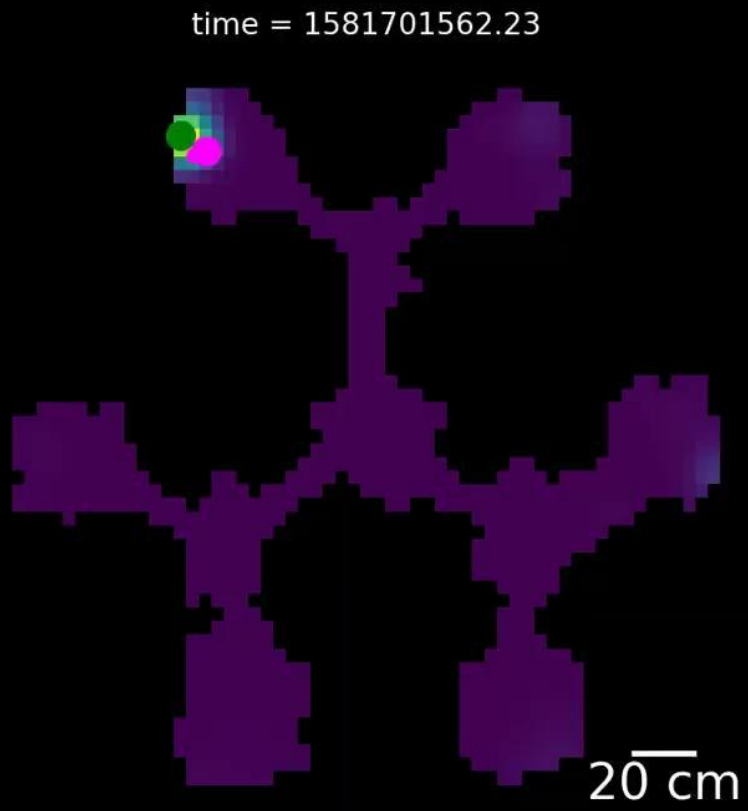
Marked Point Process Likelihood

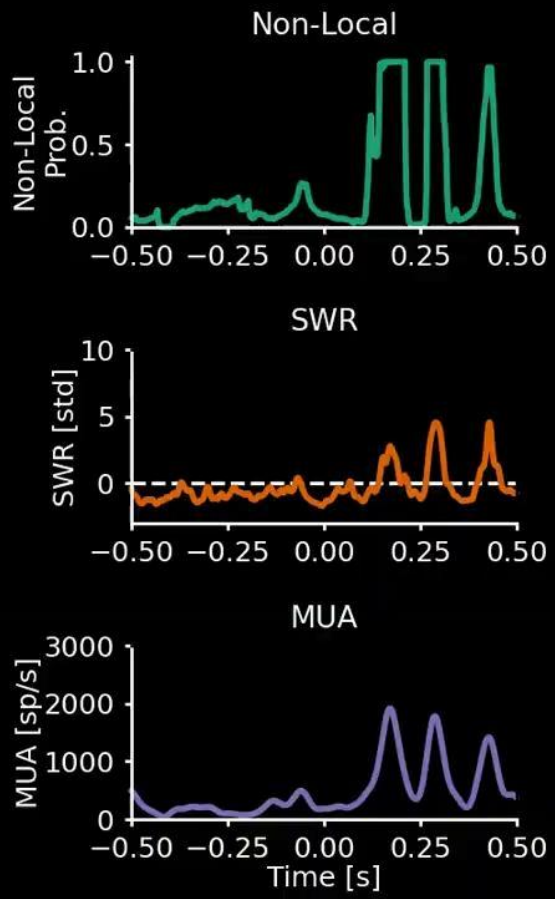
Data from Alison Comrie

Adding in a low, constant rate spiking state and fragmented state



Replay with high SWR power



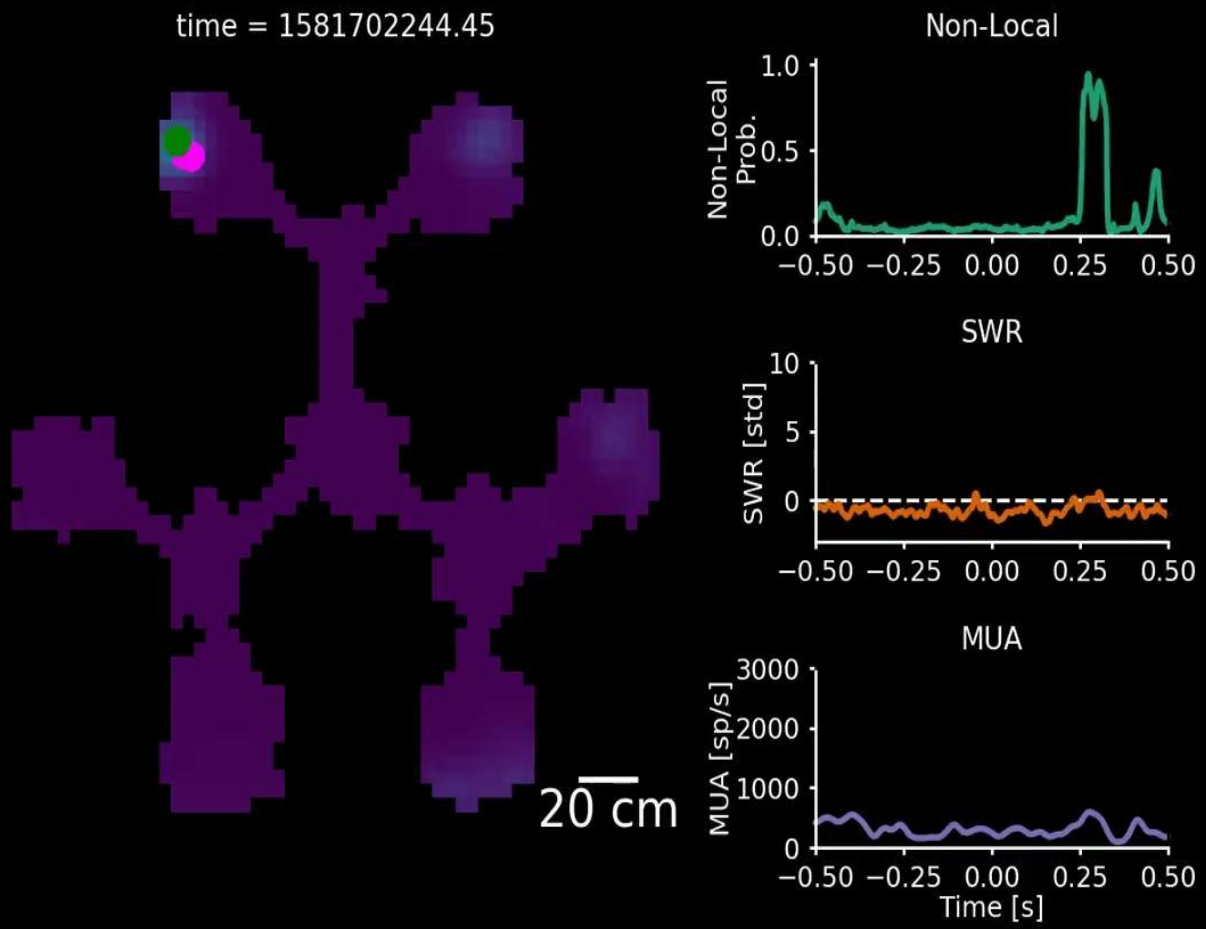


Animal's Position

Decoded Position

Data from Alison Comrie

Replay with low SWR power

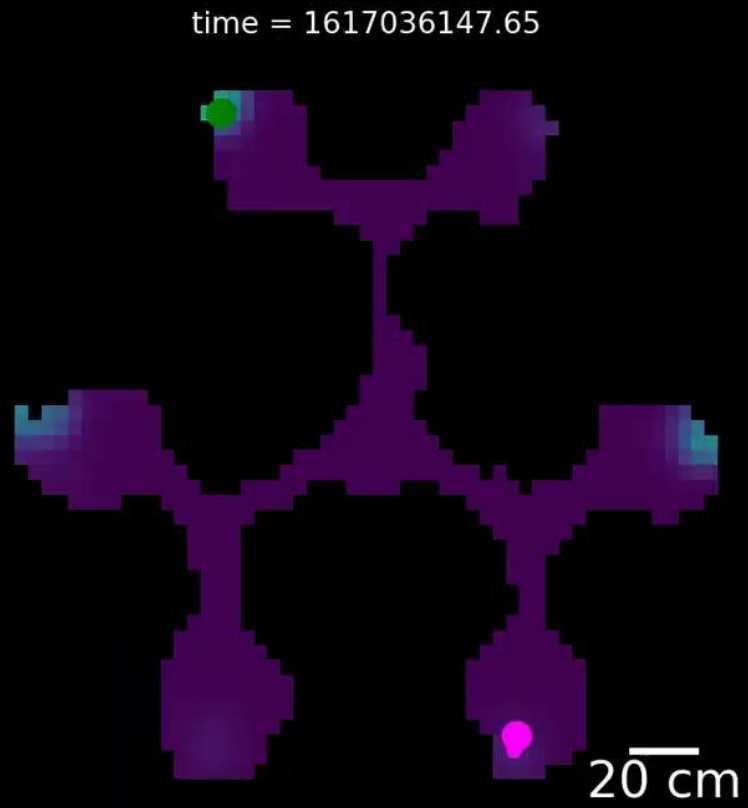


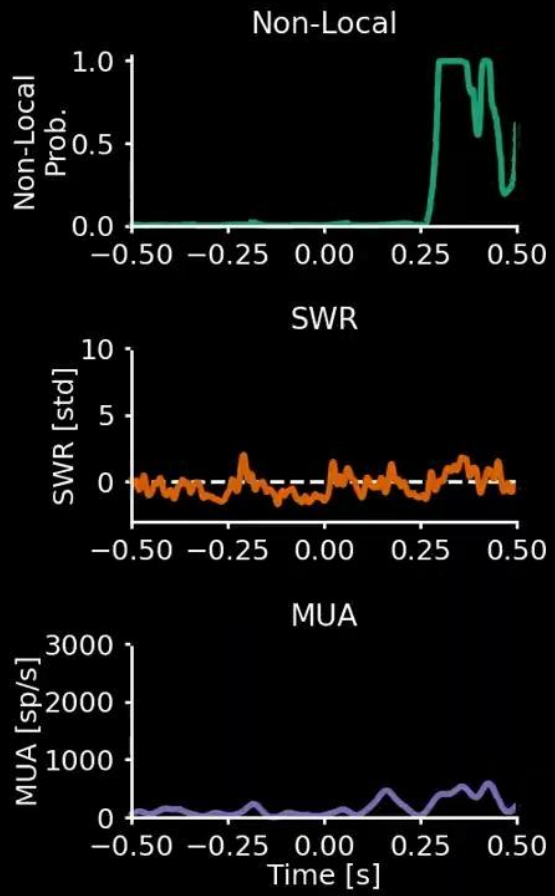
Animal's Position

Decoded Position

Data from Alison Comrie

Replay with low SWR power



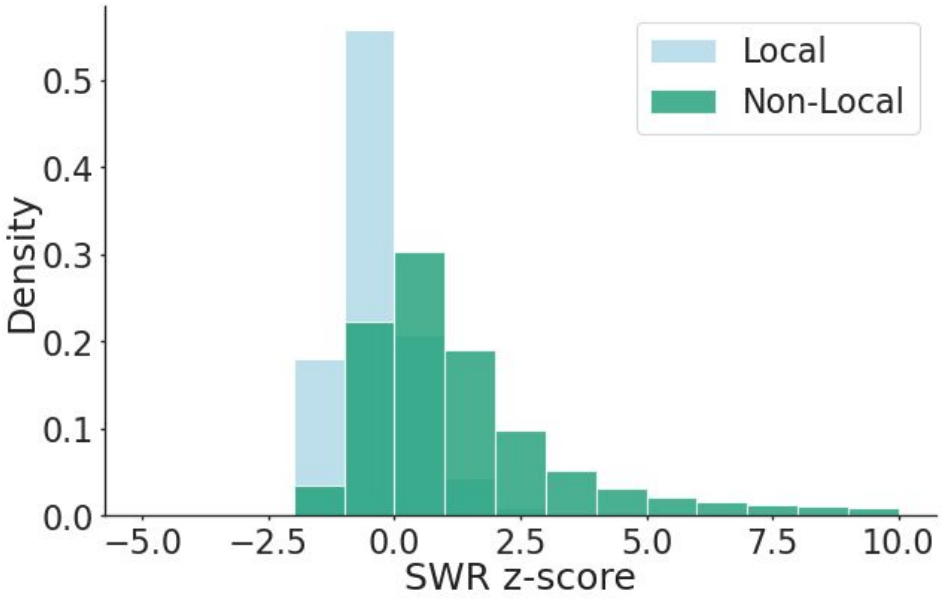


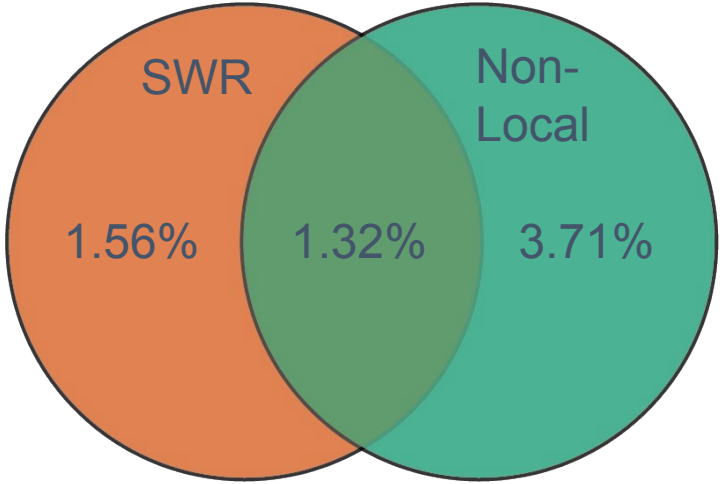
Animal's Position

Decoded Position

Data from Alison Comrie

The model can find more non-local times than previous methods





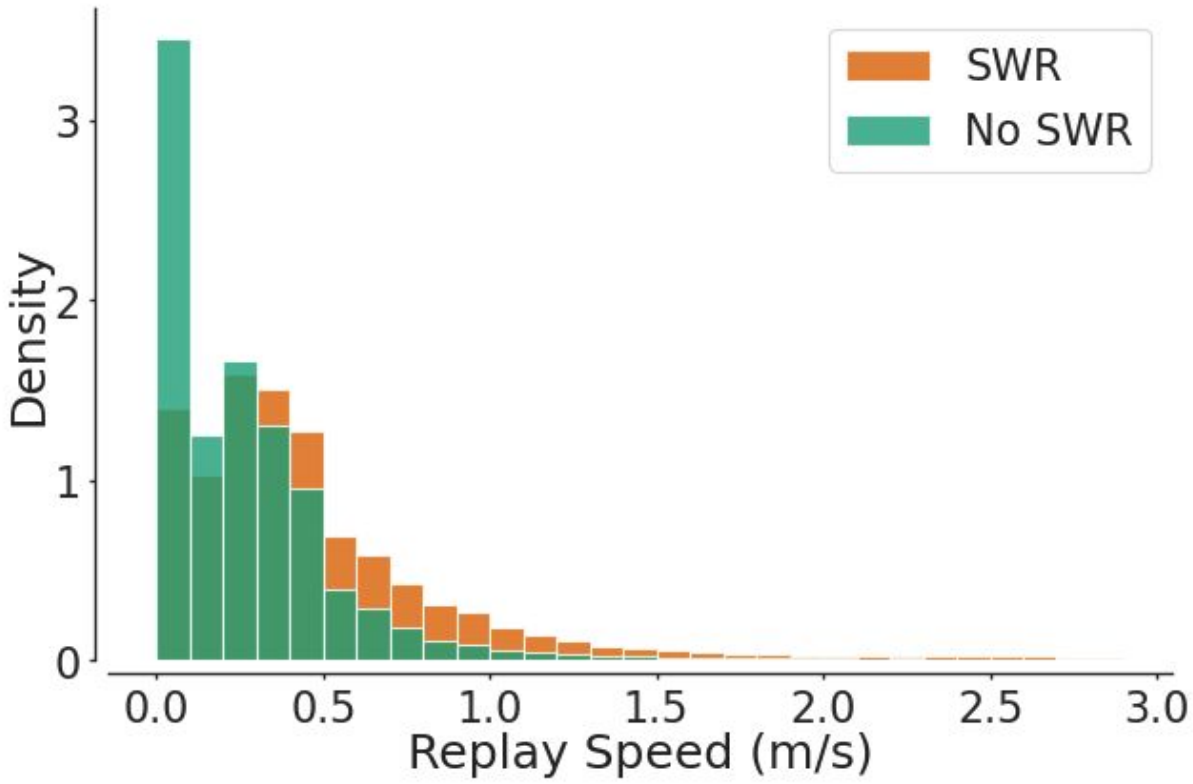
% of immobility time

+9 seconds non-local, no SWR per session (6-12 s IQR)

Data from Alison Comrie

Unpublished

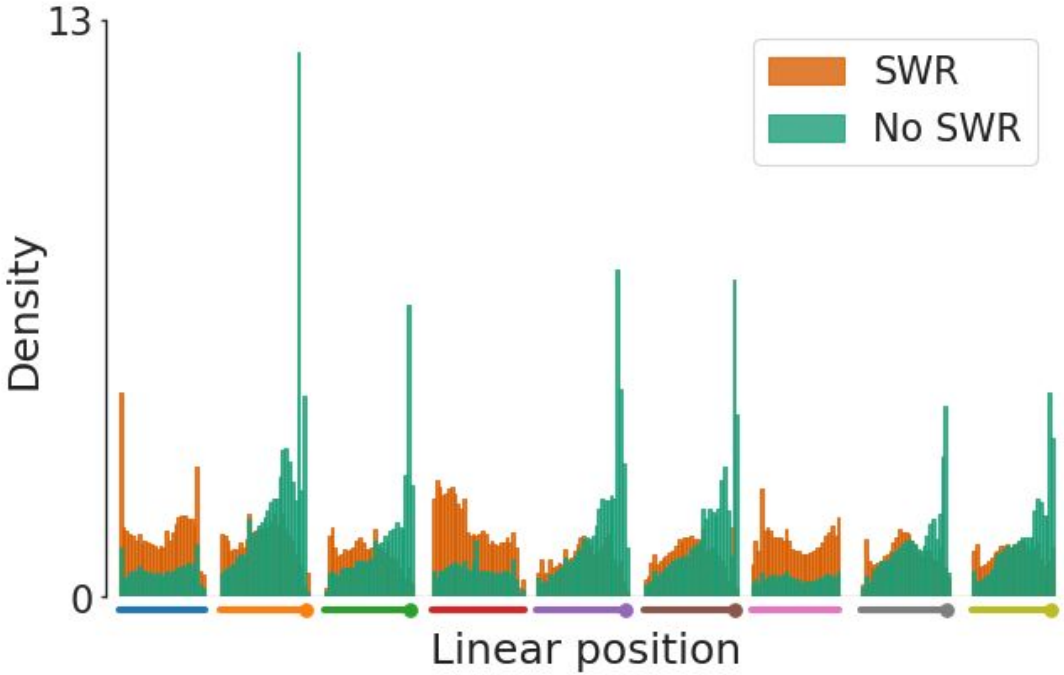
Replays with low SWR power tend move more slowly

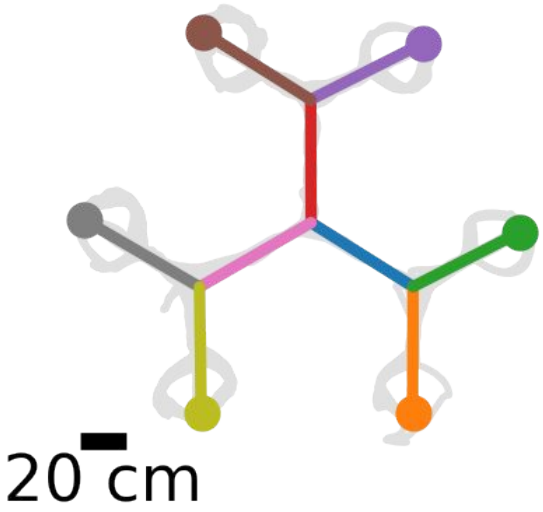


Data from Alison Comrie

Unpublished

Replay with low SWR power represents reward zones

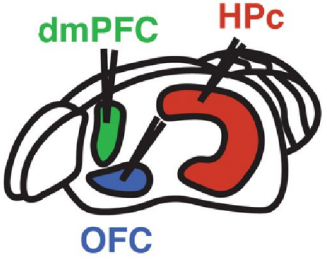


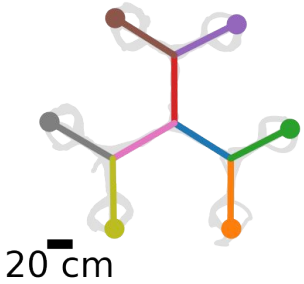


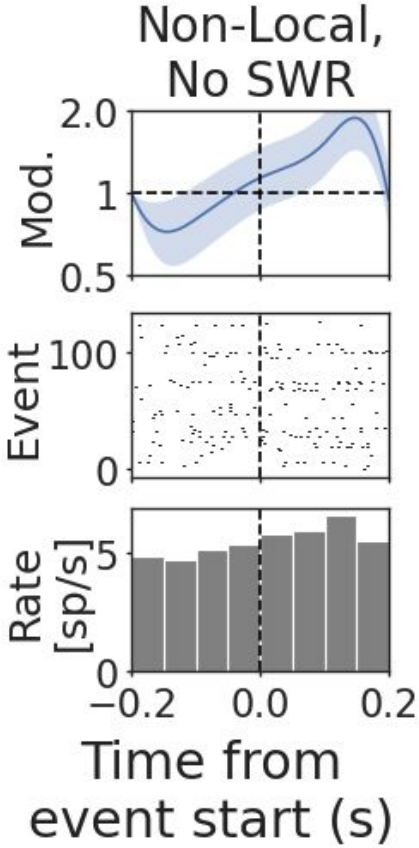
Data from Alison Comrie

Unpublished

Low SWR power replay still changes the firing rate of mPFC neurons







32% of mPFC cells (5226/16211) showed significant modulation by non-local events outside SWR

Data from Alison Comrie

Unpublished

Summary

These previously ignored, **low SWR power, slow replay events** seem to matter to the brain and may be key to understanding the function of replay.

Switching state space models have enabled us to find and characterize them with confidence.

- Exploit rich temporal structure of the data
- Using continuous and discrete latent states and defining the cognitive representation
- Separating out internal vs. external representations
- Fully utilizing the data by using clusterless waveform features
- We are starting to use them on other brain areas (Guidera…Denovellis…Frank *biorxiv* 2024)

Coulter…Denovellis…Frank *biorxiv* (2024) found events outside of SWR are more manipulable by neurofeedback

Switching state space models for real time systems for interventions
(see Chu…Denovellis…Kemere, Frank *biorxiv* 2024)

How do we assess how uncertain the model is?

We could assess something like **distance of the MAP estimate from the animal's position**.

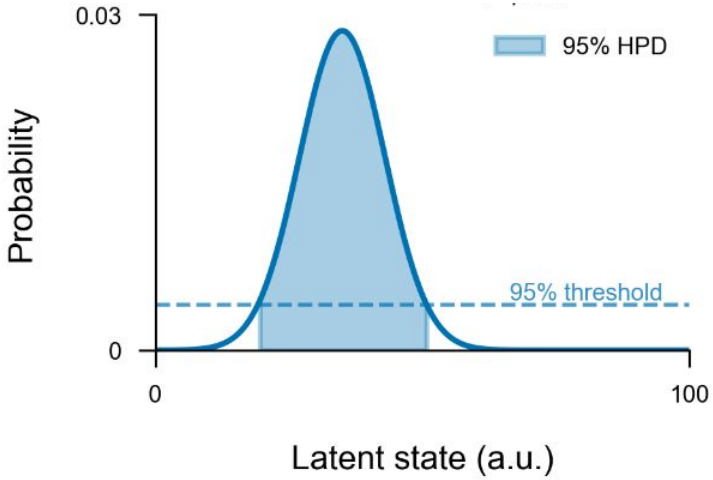
But this does not work as well when your latent state does not always represent the animal's true position (e.g. hippocampal replay, theta sequences)

How do we assess how uncertain the model is?

We can find the 95% highest values of the posterior. This is known as the **Highest Posterior Density**.

The more area this occupies, the more uncertain the model is.

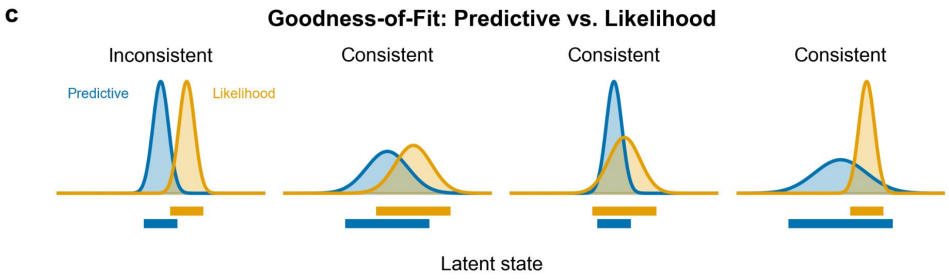
This works for multi-modal posteriors (as opposed to something like variance).

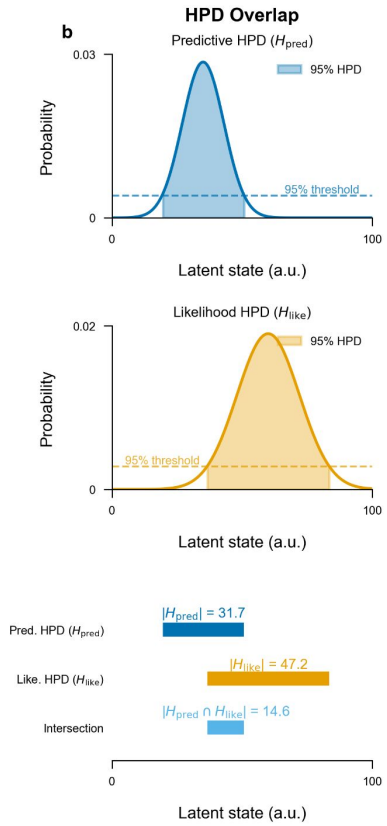


We can also check how well the data agrees with the model

Compare the prediction from the previous state with the likelihood.

If they mostly overlap, then the data and model are **consistent**.





$$\frac{|H_{pred} \cap H_{like}|}{\min(|H_{pred}|, |H_{like}|)} = \frac{14.6}{31.7} = 0.46$$

State space models vs. neural networks for decoding

Neural networks = flexible function approximators

- Deep stacks of simple nonlinear regression units
- High expressive power → can capture complex, nonlinear relationships in neural data
- Training is typically more complex (many hyperparameters, optimization details)

Interpretability vs. expressiveness

- State-space models: explicit assumptions (dynamics, observation model, priors) → easier to interpret and diagnose
- Neural networks: often less interpretable, but can model richer structure

Respecting spike statistics

- Many NN decoders treat activity as generic features
- But they *can* be made point-process aware (e.g., Poisson/NB likelihood, spike-history terms, count models, time-rescaling losses)

Computation

- State-space inference (filtering/smoothing) is often efficient and stable, and can run online
- Neural nets can be more expensive to train (backprop; large datasets), though inference can be fast once trained

Both reduce dimensionality

- Both map high-dimensional population activity → a lower-dimensional latent state (position, kinematics, cognitive variable)
- Difference: state-space does it with an explicit generative model; NNs learn the mapping end-to-end

Hybrid models are common

- Use a NN for **emissions** or **likelihood**, plus classical dynamics
- Or learn dynamics with a NN while keeping probabilistic observation models
- **Mamba**: a neural sequence model whose core computation is a learned state-space-style recurrence (efficient long-context modeling)

Decoding: a summary

Bayesian decoder is a simple, widely used algorithm in neuroscience (likelihood × prior → posterior).

Decoded results depend on user choices (binning, smoothing, priors, encoding assumptions, posterior readout, etc.)

State-space models incorporate dynamical assumptions principledly (via a transition model) without relying on ad hoc smoothing windows.

Switching state-space models add flexibility by allowing multiple dynamical "modes" (e.g., stationary vs moving, forward vs reverse, different sequence types)

Model checking is as important for decoding models as encoding models

The course: a summary

- Understanding spatial navigation and neuroscience requires knowledge in a wide range of fields.
- This course touched briefly on a lot of things (anatomy, systems neuroscience, point processes, statistics, signal processing, decoding). It's important to know how much you do and don't know and what to look up to understand more.
- It's important to dig deep with your tools. Neuroscience begins with data collection, but the analysis (and coding) is a large part of doing science. You need to make sure you refine those skills and not neglect them.
- At the same time, it's important to leverage prior and deep contextual knowledge to inform your analysis. Engage critically with statistics.
- Do the simple thing first and plot your data.